

This document contains the proposed syllabus for ECE, 3rd semester. The assessment and rubrics for Mathematics and Community Project (Project Based Learning) Societal Project courses are provided along with the individual syllabus. The assessment and rubrics for all other subjects (PCC, IPCC, PCC-L and AEC) are provided at the end of this file. The rubrics provided are suggested by the BoS, however individual departments may modify them as required.

Students are expected to perform SL/TW as part of their course evaluation. Individual institutions may follow their own structure for this. The below is the suggestion from ECE BoS for SL/TW for Network Analysis course (3 credit course), faculty may adopt this for other courses or modify as required. Other components can be considered as required, e.g. quiz, course-project, seminar report etc.

Self-Learning component for Assignment:

Evaluation Method	Self-Learning Topics	COs addressed	Duration of self-learning study	Total Duration of Self learning study
Assignment (10 marks) Covering Modules 1,2,3 Students are expected to submit a report which will be evaluated	Concepts of Sensor loading	1,2	8 hrs	28 hrs
	Applications of Superposition and Thevenin's theorem	1,2,3	10 hrs	
	Impact of phase delay and phase lag in circuit design	1,2,3	10 hrs	

Self-Learning component for Quiz:

Evaluation Method	Self-Learning Topics	COs addressed	Duration of self-learning study	Total Duration of Self learning study
Quiz (10 marks) Covering Module 3,4 Students will take the quiz and evaluation to be done accordingly. The hours given for the self learning indicate the preparation time for the quiz.	Time domain solutions of RLC circuits	3,4	8 hrs	20 hrs
	Circuit representations using Laplace transforms	3,4	7 hrs	
	Waveform synthesis	4	5 hrs	

Transform Techniques and Optimization Theory			
Course Code	1BMATEC301	Scheme	2025
Type of Course	ASC	Semester	III
Teaching Hours/Week (L:T:P)	3:2:0	CIE Marks	50
Total Hours of Pedagogy (L :T:P:SL&TW)	42:28:0:50	SEE Marks	50
Credits	4	Total Marks	100
Examination type (SEE)	Theory	Exam Hours	3
Course outcome (Course Skill Set) At the end of the course, the student will be able to: 1. Develop Fourier series representations of periodic functions and perform harmonic analysis for functions with different periodicities, discontinuities, and symmetry properties. 2. Apply Fourier transforms, including sine, cosine, discrete Fourier transforms (DFT), and Fast Fourier Transforms (FFT), to analyze signals and solve engineering problems involving differential equations. 3. Utilize Z-transforms, their properties, inverse transforms, and convolution techniques to analyze discrete-time systems and solve difference equations. 4. Formulate and solve Linear Programming Problems using graphical, simplex, and Big-M methods to obtain optimal solutions in engineering and management applications. 5. Apply transportation and assignment models, including optimization techniques such as Vogel's Approximation Method and Hungarian Method, to solve resource allocation and decision-making problems.			
Module-1			
Fourier series: Periodic functions, Euler's Formula, Conditions for Fourier expansion (Dirichlet's condition), Fourier series of functions i) having points of discontinuity, ii) with period 2π , iii) with change of interval (arbitrary period) and iv) Even and odd functions. Half rang Fourier series. Practical harmonic analysis.			
			Number of Hours:13
Module-2			
Fourier Transforms: Infinite Fourier transform, Infinite Fourier sine and cosine transforms, Properties-Linear property, Change of scale, shifting property, Modulation theorem. Inverse transforms, Convolution theorem and its significance, Fourier transform of derivatives and integrals, Solution of PDE's using Fourier transforms. Discrete Fourier transform (DFT), Inverse DFT, Fast Fourier transform (FFT)			
			Number of Hours:15
Module-3			
Discrete Transforms - Z transforms: Z-transform of Standard functions, Linearity property, Damping rule, Shifting property, Initial and final value theorems, Inverse Z-transform- Power series method, Partial fraction method, inverse integral method. Convolution theorem, Application of Z-transforms to solve difference equations.			
			Number of Hours:15
Module-4			
Linear programming problems: Introduction to optimization techniques, nature and characteristics of operation research; Introduction to Linear programming, graphical solution, solution by simplex and Big M method.			
			Number of Hours:13
Module-5			
Transportation problems: Formulation of transportation problem, Initial Basic feasible solution by North-West Corner method, Least cost method, Vogel's approximation method. Optimal solutions-Problems. Formulation of assignment problems, Hungarian method-Problems			
			Number of Hours:14

Suggested Learning Resources: (Text Book/ Reference Book/ Manuals):**Text books:**

1. B.S. Grewal, Higher Engineering Mathematics, 45th Edition, Khanna Publishers.
2. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley India.
3. H.K. Dass and Er. Rajnish Verma, Higher Engineering Mathematics, S. Chand Publications.

Reference books / Manuals:

1. Ruel V. Churchill and James Ward Brown, Fourier Series and Boundary Value Problems, McGraw-Hill Education.
2. Hamdy A. Taha, Operations Research: An Introduction, Pearson Education.
3. Schaum's Outline of Fourier Analysis and Applications, McGraw-Hill.
4. Murray R. Spiegel, John Schiller and R. Alu Srinivasan, Schaum's Outline of Probability and Statistics, McGraw-Hill.
5. S. D. Sharma, Operations Research, Kedar Nath Ram Nath Publications.

Web links and Video Lectures (e-Resources):

- NPTEL Courses on Engineering Mathematics
- MIT Open Course Ware Mathematics
- SWAYAM Online Courses

Teaching-Learning Process (Innovative Delivery Methods):

The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes.

1. **Flipped Classroom Approach**
 - Students review video lectures and reading materials before class.
 - Classroom sessions focus on problem-solving, discussions, and applications of transform techniques and optimization models.
2. **Problem-Based Learning (PBL)**
 - Real-world engineering problems involving Fourier Transforms, Z-Transforms, and Linear Programming are assigned for collaborative solution development.
3. **ICT-Enabled Teaching** (may be conducted in Tutorial hours)
 - Use of MATLAB, Python, and online simulation tools to visualize transforms, optimization problems, and signal-processing applications.
4. **Collaborative Learning** (may be conducted in Tutorial hours)
 - Group assignments and mini-projects on optimization models and transform applications.

Assessment Structure:

The assessment in each course is divided equally between Continuous Internal Evaluation (CIE) and the Semester End Examination (SEE), with each carrying 50% weightage.

- To qualify and become eligible to appear for SEE, in the **CIE**, a student must score at least **40% of 50 marks**, i.e., **20 marks**.
- To pass the **SEE**, a student must score at least **35% of 50 marks**, i.e., **18 marks**.
- Notwithstanding the above, a student is considered to have **passed the course**, provided the combined total of **CIE and SEE** is at least **40 out of 100 marks**.

Continuous Comprehensive Assessments (CCA)+ Internal Assessment Test (IA) = Continuous Internal Evaluation CIE [25+25=50 marks]

A minimum of **two Internal Assessment Tests (IA)** shall be conducted, carrying a total of **25 marks**.

In addition, **Continuous Comprehensive Assessment (CCA)** shall be conducted for a total of **25 marks**.

It is recommended to include a **maximum of two learning activities** as part of the CCA to foster the **holistic development of students**. These activities shall be:

Sl. No.	Name of the Learning activity	Maximum Marks
1.	Assignment (at RBL3 levels)	10
2.	GATE-based Aptitude Test	15

Rubrics for Learning Activity-1: Assignment (10 Marks)

Performance Indicator	Superior (10)	Good (8)	Fair (6)	Needs Improvement (5)
Understanding of Concepts	Demonstrates excellent conceptual clarity and application	Good understanding with minor errors	Basic understanding	Poor understanding
Problem Solving Ability	Solves all problems correctly with proper methodology	Solves most problems correctly	Partial solutions with errors	Unable to solve problems effectively
Mathematical Accuracy	Calculations and derivations are error-free	Minor computational errors	Several errors	Significant errors
Presentation & Organization	Well-structured, neat, and professionally presented	Organized with minor deficiencies	Average presentation	Poor presentation

Rubrics for Learning Activity-2: GATE-Based Aptitude Test (15 Marks)

Performance Indicator	Superior (15)	Good (12)	Fair (10)	Needs Improvement (8)
Quantitative Aptitude	≥90% correct responses	75–89% correct	60–74% correct	<60% correct
Analytical Thinking	Excellent logical reasoning and interpretation	Good reasoning ability	Moderate reasoning	Weak reasoning
Time Management	Completes all questions within allotted time	Completes most questions	Requires additional time	Unable to complete
Accuracy & Precision	Very high accuracy	Good accuracy	Moderate accuracy	Low accuracy

Suggested Learning Activities may include (but are not limited to):

- Course Project
- Case Study Presentation
- Programming Assignment
- Tool/Software Exploration
- Literature Review
- Open Book Test (preferably at RBL4 and RBL5 levels)
- GATE-based Aptitude Test
- Assignment (at RBL3, RBL4, or RBL5 levels)
- Any other relevant and innovative academic activity

Suggested Innovative Delivery Methods may include (but are not limited to):		
• Flipped Classroom • Problem-Based Learning (PBL) • Case-Based Teaching • Simulation and Virtual Labs • Partial Delivery of course by Industry expert/ industrial visits • ICT-Enabled Teaching • Role Play		
Term Work (TW) and Self Learning (SL) components in Number of Hours per semester		
Term work (includes assignments, seminars, micro projects, industrial visits, any other student activities etc.)		
Sl. No.	Term Work (TW) Activity	Number of Hours / Semester
1.	Seminar on Fourier Series of periodic functions and harmonic analysis	5
2.	MATLAB/Python Simulation of Fourier and Z-Transform problems	5
3.	Mini Project on Linear Programming Models in Engineering Applications	5
4.	Case Study on Signal Processing using Transform Techniques	5
SL: Self Learning, MOOCs, spoken tutorials, online educational resources etc.		
Sl. No.	Self-Learning (SL) Activity	Number of Hours / Semester
1.	Application of Fourier Series in signal representation and on harmonic analysis of periodic signals.	6
2.	Explore applications of DFT and FFT in digital signal processing.	6
3.	Solve a set of difference equations using Z-transforms	6
4.	Develop and solve two real-world optimization problems	6
5.	Study industrial applications of transportation and assignment models	6

Digital System Design Using Verilog			
Course Code	1BEC302	Scheme	2025
Type of course	IPCC	Semester	III
Teaching Hours/Week (L:T:P)	3:0:2	CIE Marks	50
Total Hours of Pedagogy (L:T:P:SL&TW)	42:0:28:50:0	SEE Marks	50
Credits	4	Total Marks	100
Type of Examination (IPCC):	Theory- CIE +SEE, Lab- CIE only.	Exam Hours	3
Course outcome (Course Skill Set) At the end of the course, students will be able to: 1. Apply Boolean algebra principles to simplify logic functions using Karnaugh Maps (K-Maps) and the Quine–McCluskey minimization technique. 2. Analyze the operation and behavior of combinational logic circuits and standard logic building blocks. 3. Design combinational digital systems using logic gates, multiplexers, decoders, and arithmetic circuits. 4. Analyze and design synchronous and asynchronous sequential circuits using SR, JK, D, and T flip-flops. 5. Model, simulate, and verify combinational and sequential digital circuits using the Verilog Hardware Description Language (HDL).			
Module-1			
Principles of Combinational Logic: Definition of combinational logic, Canonical forms, Generation of switching equations from truth tables, Karnaugh maps-up to 4 variables, Quine McCluskey Minimization Technique. Quine-McCluskey using Don't Care Terms. Text book 1: Section 3.1 to 3.5 Number of Hours: 8			
Module-2			
Logic Design with MSI Components and Programmable Logic Devices: Binary Adders and Subtractors, Decimal Adders, Comparators, Decoders, Encoders, Multiplexers, Programmable Logic Devices (PLDs) Text book 2: Section 5.1 to 5.7 Number of Hours: 8			
Module-3			
Introduction to Verilog: Structure of Verilog module, Styles of Description, Ports, Operators, Data Types, Text book 3: Section 1.1 to 1.6.2 (only Verilog) Verilog Data flow description: Highlights of Data flow description, Signal Declaration and Assignment Statement, Structure of Data flow description. Text book 3: Section 2.1 to 2.2 (only Verilog) Number of Hours: 8			
Module-4			
Flip-Flops and its Applications: The Master-Slave Flip-flops (Pulse-Triggered flip-flops): The basic bistable element, Latches, Timing Considerations, SR flip flops, JK flip flops, Characteristic equations, Registers, Binary Ripple Counters, Synchronous Binary Counters, Counters based on Shift Registers, Design of Synchronous mod-n Counter using clocked T, J K, D and SR flip-flops. Text book 2: Section 6.1 to 6.4, 6.6 to 6.9 (Excluding 6.9.3). Number of Hours: 10			
Module-5			
Verilog Behavioral Description: Structure, Variable Assignment Statement, Sequential Statements, Loop Statements, Verilog Behavioral Description of Multiplexers. Text book 3: Section 3.1 to 3.4 (only Verilog) Verilog Structural Description: Highlights of Structural description, Organization of structural description, Structural description of ripple carry adder. Text book 3: Section 4.1 to 4.2 Number of Hours: 10			

PRACTICAL COMPONENTS OF IPCC

Open-source tool /Siemens HyperLynx AMS/Xilinx can be used for simulation.

1. Boolean Expression Simplification and Logic Circuit Implementation

- a) Simplify the given Boolean expression.
- b) Design a Full Adder.

Realize the corresponding circuits using a breadboard or digital trainer kit and verify their operation.

2. Adder/Subtractor Circuits Using Verilog Dataflow Description

- a) Realize a Half Adder.
- b) Realize a Full Adder.
- c) Realize a Half Subtractor.
- d) Realize a Full Subtractor.

Implement the circuits using Verilog dataflow description. Simulate the designs and test them on an FPGA board.

3. 4-bit ALU Using Verilog

Write Verilog code for a 4-bit Arithmetic Logic Unit (ALU). Simulate the design and test it on an FPGA board.

4. Code Converters Using Verilog Behavioral Description

Write Verilog code for the following code converters using behavioral description. Simulate the designs and test them on an FPGA board.

- a) Gray-to-Binary and Binary-to-Gray
- b) Binary-to-Excess-3 and Excess-3-to-Binary

5. Multiplexer, Decoder and Encoder Using Verilog

Realize the following using Verilog behavioral description. Simulate the designs and test them on an FPGA board:

- a) 8:1 Multiplexer
- b) 3:8 Decoder
- c) 8:3 Priority Encoder

6. BCD-to-7-Segment Display and Switch/LED Interfacing Using Verilog

- a) Write a Verilog program to design a BCD-to-7-segment display decoder.
 - b) Write a Verilog program to interface switches and LEDs with an FPGA/CPLD and demonstrate their operation.
- Simulate the designs and test them on an FPGA board.

7. Demultiplexer and Comparators Using Verilog

Realize the following using Verilog behavioral description. Simulate the designs and test them on an FPGA board:

- a) 1:8 Demultiplexer
- b) 2-bit Comparator
- c) 4-bit Magnitude Comparator

8. Flip-Flops Using Verilog

Realize the following flip-flops using Verilog behavioral description. Simulate the designs and test them on an FPGA board:

- a) JK Flip-Flop
- b) SR Flip-Flop
- c) T Flip-Flop

d) D Flip-Flop

9. Up/Down Counters Using Verilog Structural Description

Design and implement BCD and binary up/down counters using Verilog structural description. Simulate the designs and test them on an FPGA board.

10. Stepper Motor Interfacing Using Verilog

Write a Verilog program to interface a stepper motor with an FPGA/CPLD through a suitable motor driver circuit and rotate the motor in the specified direction by a specified number of steps.

11. FSM-Based Sequence Detector Using Verilog

Design a Finite State Machine (FSM) to detect a given sequence, such as 1010. The input shall be applied serially, and the output shall go HIGH whenever the sequence is detected. The design shall support overlapping sequences. Write the Verilog code, simulate the design using a testbench, and test it on an FPGA board.

12. Sequential Circuit Design Using Flip-Flops

Design a specified sequential circuit using flip-flops, write the corresponding Verilog code, simulate the design using a testbench, and test it on an FPGA board.

Note: Experiment 1 focuses on hardware-based logic circuit realization using a breadboard or digital trainer kit. Experiments 2–12 cover combinational circuits, code converters, sequential circuits, counters, FPGA interfacing, display interfacing, motor interfacing, and FSM design using Verilog HDL.

Suggested Learning Resources: (Textbook/ Reference Book/ Manuals):

Text books:

1. J. M. Yarbrough, Digital Logic Applications and Design. Clifton Park, NY, USA: Thomson Learning, 2006.
2. D. D. Givone, Digital Principles and Design. New York, NY, USA: McGraw-Hill, 2002.
3. N. Botros, HDL with Digital Design: VHDL and Verilog. Dulles, VA, USA: Mercury Learning and Information, 2015

Reference books / Manuals:

1. Fundamentals of logic design, by Charles H Roth Jr., Cengage Learning
2. Fundamentals of HDL, by Cyril PR, Pearson/Sanguine 2010
3. Logic Design, by Sudhakar Samuel, Pearson/Sanguine, 2007

Web links and Video Lectures (e-Resources):

- www.chipverify.com
- <https://www.youtube.com/watch?v=vHLBO05TeyU&list=PLwdnzlV3ogoVIY7iVqr-FhWUQEX7JDdiP>
- <https://www.youtube.com/playlist?list=PLbRMhDVUMngePP5JcezxImF-FzOC9wstz>

Teaching-Learning Process (Innovative Delivery Methods):

The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes.

1. Interactive lectures using digital design visualization tools and multimedia presentations to explain logic design concepts.
2. Simulation-based learning using Verilog HDL simulators to model, simulate, and verify digital circuits.
3. Flipped classroom methodology, where students study digital design concepts before class and engage in design exercises during class.
4. Mini-projects and project-based learning involving the design and verification of digital systems such as counters, ALUs, finite state machines, and controllers.
5. Collaborative learning activities through group discussions, peer reviews, and team-based design projects.
6. Continuous assessment through quizzes, Verilog coding challenges, and design reviews to reinforce learning and provide timely feedback.

Questions from the practical component of the IPCC subject shall not appear in the theory SEE component.

Network Analysis			
Course Code	1BEC303	Scheme	2025
Type of Course	PCC	Semester	III
Teaching Hours/Week (L:T:P: S)	3:2:0	CIE Marks	50
Total Hours of Pedagogy (L :T:P:SL&TW)	42:28:0:50	SEE Marks	50
Credits	4	Total Marks	100
Examination type (SEE)	Theory	Exam Hours	3
Course outcome (Course Skill Set) At the end of the course, the student will be able to: 1. Analyze electronic networks using mesh and nodal analysis techniques for DC and AC circuits with dependent and independent sources. 2. Evaluate electronic circuit responses (currents, voltages, and power) using network theorems. 3. Analyze RL, RC, and RLC circuits under transient conditions using initial and final condition concepts. 4. Analyze linear electronic circuits using Laplace transform techniques. 5. Evaluate the performance of two-port networks by determining parameters and impedance characteristics.			
Module-1			
Basic Concepts- Mesh and Node Analysis: Introduction, Source of Electrical Energy, Kirchoffs Laws, Source transformations, General Network Transformations, Mesh and Node analysis with linearly dependent and independent sources for DC and AC networks. Text book 1: 1.1, 3.2, 3.3, 4.1-4.4, 5.2, 5.5, 10.1-10.6 Number of Hours: 15			
Module-2			
Network Theorems for DC and AC circuits: Introduction, Superposition Theorem, Reciprocity Theorem, Conditions for Reciprocity, Examples of Reciprocal/Non-Reciprocal circuits, Thevenin's Theorem, Maximum Power transfer theorem. Case Study-1: Maximum Power Transfer in Electronic Systems Design. Text book 1: 5.1, 5.3,5.4,10.7; Textbook 2: 4.9 Number of Hours: 13			
Module-3			
Transient behavior and initial conditions: Behavior of circuit elements under switching condition and their Representation, Evaluation of initial and final conditions in RL, RC and RLC circuits for AC and DC excitations. Case Study: RC Circuit-Based Power-ON Delay in Electronic Indicator Systems. Text book 1: 7.1, 7.2, 7.7, 8.1 – 8.9, 9.1-9.7, 10.1-10.7 Number of Hours: 14			
Module-4			
Laplace Transformation & Applications: Solution of networks, step, ramp and impulse responses, Waveform Synthesis, Transfer function of elementary passive circuits, Network Synthesis Case Study: Concept of poles for first order passive filters and their use in computing time response characteristics and bandwidth. Text book 1: 14.3 – 14.7, 15.1-15.8; Textbook 2: 6.8. Number of Hours: 14			
Module-5			
Two -Port Network: Introduction, Characterisation of Linear Time-invariant Two-port Networks, Open- Circuit Impedance Parameters, Short- circuit Admittance Parameters, Transmission Parameters, Short introduction to Hybrid parameters (1 hour material), Why multiple representations are required, Two-port Symmetry, Input impedance in terms of Two-port Parameters, Output Impedance. Text book 1: 17.1 – 17.6 Number of Hours: 14			

Suggested Learning Resources: (Text Book/ Reference Book/ Manuals):**Text books:**

1. Hayt, Kemmerly and Durbin-Engineering Circuit Analysis, TMH 8th Edition, 2012
2. Network Analysis – A Simplified Approach, K Channa Venkatesh and D Ganesh Rao, Cengage India Private Limited, 2012, Second Edition, ISBN-13: 978-8131533215

Reference books / Manuals:

1. M.E.Van Valkenburg (2000), Network Analysis, Prentice Hall of India, 3rd edition, 2000, ISBN:9780136110958.
2. J.David Irwin/ R.Mark Nelms- Basic Engineering Circuit Analysis JohnWiley,8th ed,2006.
3. Charles K Alexander and Mathew NO Sadiku-Fundamentals of Electric Circuits, Tata Page 17 BE III Semester in ECE/TCE 15.09.2023 McGraw-Hill,3rd Ed,2009.

Web links and Video Lectures (e-Resources):

- <https://nptel.ac.in/courses/108105159>

Teaching-Learning Process (Innovative Delivery Methods):

The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes.

1. Ensure at least 2 hours of review of mathematical concepts such as matrix algebra, complex variables, first and second order differential equations during the tutorial sessions
2. Present case studies for each module which have industry relevance
3. Use of simulation tool or even hardware demonstrations in the classroom to illustrate complex concepts

Questions from the Case Study topics shall not appear in the theory SEE component

Analog Electronics and Linear Integrated Circuits			
Course Code	1BEC304	Scheme	2025
Type of Course	PCC	Semester	III
Teaching Hours/Week (L:T:P)	3:0:0	CIE Marks	50
Total Hours of Pedagogy (L :T:P:SL&TW)	42:0:0:48	SEE Marks	50
Credits	3	Total Marks	100
Examination type (SEE)	Theory	Exam Hours	3
Course outcome (Course Skill Set) At the end of the course, the student will be able to: 1. Describe the working principle, fundamental characteristics of various semiconductor devices such as transistors, MOSFETs, and operational amplifiers in basic electronic circuits. 2. Analyze basic amplifier circuits using the principles BJTs, MOSFETS, and operational amplifiers. 3. Illustrate the fundamental concepts of oscillators and timers. 4. Design basic filters and power amplifiers. 5. Analyze the fundamental concepts of linear Op-Amps, linear Op-Amps and their applications. 6. Analyze the electronic circuit schematic and its working.			
Module-1			
BJT AC Models: Base Biased Amplifier, Emitter Biased Amplifier, Small Signal Operation, AC Beta, AC Resistance of The Emitter Diode, Two Transistor Models, Analyzing Amplifier. Voltage Amplifiers: Voltage Gain, Multistage Amplifiers. CC And CB Amplifiers: CC Amplifier, Output Impedance, Cascading CE And CC, Darlington Connections, Voltage Regulation, The Common Base Amplifier. Text book 1: 8.1-8.7, 8.9, 9.1, 9.3-9.8			
			Number of Hours:9
Module-2			
MOSFET Biasing in MOS Amplifier Circuits: Fixing VGS, Fixing VG, Drain to Gate Feedback Resistor. Small Signal Operation and Modelling: The DC Bias Point, Signal Current in Drain, Voltage Gain, Small Signal Equivalent Circuit Models, Transconductance, The T Equivalent Circuit Model. MOSFET Amplifier Configuration: Basic Configurations, Characterizing Amplifiers, CS Amplifier with And Without Source Resistance, The Common Gate Amplifier, Source Follower. Text book 2: 7.2, 7.2.1, 7.3, 7.3.1 - 7.3.6, 7.4, 7.4.1			
			Number of Hours:9
Module-3			
Negative Feedback: Four Types of Negative Feedback, VCVS Voltage Gain, Other VCVS Equations, ICVS Amplifier, VCIS Amplifier, ICIS Amplifier (No Mathematical Derivation). Oscillator: Theory of Sinusoidal Oscillation, The Wein-Bridge Oscillator, RC Phase Shift Oscillator, The Colpitts Oscillator, Hartley Oscillator, Crystal Oscillator. The 555 Timer: Monostable Operation, Astable Operation. Text book 1: 17.1-17.6, 21.1-21.8			
			Number of Hours:8
Module-4			
Power Amplifiers: Amplifier Terms, Two Load Lines, Class A Operation, Class B Operation, Class B Push Pull Emitter Follower, Class C Operation. Active Filters: Ideal Responses, First Order Stages, VCVS Unity Gain Second Order Low Pass Filters, VCVS Equal Component Low Pass Filters, VCVS High Pass Filters, MFB Bandpass Filters, Bandstop Filters. Text book 1: 10.1-10.5, 10.8, 19.1, 19.4, 19.5, 19.7-19.10			
			Number of Hours:8

Module-5	
Linear Op-amp Circuits: Digital to Analog Conversion: Weighted Resistor DAC, R-2R DAC, Analog to Digital Conversion: Digital RAMP ADC, Successive Approximation ADC. Signal Processing Circuits: Saturating and Non Saturating Precision HWR, Precision FWR Using HWR and Summing Circuit. Nonlinear Op-Amp Circuits: Zero Crossing Detector, Inverting Schmitt Trigger Circuit, Non-Inverting Schmitt Trigger Circuit. DC Voltage Regulators: Introduction, Voltage Regulator Basics. Text book 3: 15.2, 15.4, 9.1, 9.2, 8.2-8.4, 13.1	
	Number of Hours:8
Suggested Learning Resources: (Text Book/ Reference Book/ Manuals): Text books: 1. Albert Malvino, David J Bates, Patrick E. Hoppe, Electronic Principles, 9th Edition, Mc Graw Hill Education, 2021. 2. Adel S Sedra, Kenneth C Smith, Microelectronic Circuits, 7th Edition, Oxford University Press, 2015. 3. David A. Bell, Operational Amplifiers and Linear ICs, 3rd Edition, Oxford University Press, 2011. Reference books / Manuals: 1. Jacob Millman, Christos C. Halkias, Integrated Electronics: Analog and Digital Circuits and Systems, McGraw-Hill, 2015. 2. Boylestad & Nashelsky, Electronic Devices and Circuit, Eleventh Edition, Pearson, January 2015.	
Web links and Video Lectures (e-Resources): • https://nptel.ac.in/courses/108102112 • https://onlinecourses.nptel.ac.in/noc25_ee157/preview • https://nptel.ac.in/courses/108106188	
Teaching-Learning Process (Innovative Delivery Methods): The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes. 1. Lecture method (L) does not mean only the traditional lecture method, but a different type of teaching method may be adopted to develop the outcomes. 2. Show Video/animation films to explain the functioning of various analog and digital circuits. 3. Adopt Problem Based Learning (PBL), which fosters students' Analytical skills, develop thinking skills such as the ability to evaluate, generalize, and analyse information rather than simply recall it. 4. Show the different ways to solve the same problem and encourage the students to come up with their own creative ways to solve them. 5. Arrange visits to nearby industries to give brief information about the electronics manufacturing industry. 6. Discuss how every concept can be applied to the real world - and when that's possible, it helps improve the students' understanding.	

Python Programming			
Course Code	1BEC305	Scheme	2025
Type of Course	PCC	Semester	III
Teaching Hours/Week (L:T:P)	3:0:0	CIE Marks	50
Total Hours of Pedagogy (L :T:P:SL&TW)	42:0:0:48	SEE Marks	50
Credits	3	Total Marks	100
Examination type (SEE)	Theory	Exam Hours	3
Course outcome (Course Skill Set) At the end of the course, the student will be able to: 1. Acquire programming skills in Python 2. Demonstrate data structure representation using Python 3. Develop the skill of pattern matching and files in Python 4. Acquire Object Oriented Skills in Python 5. Develop the ability to write database applications in Python			
Module-1			
Python Basics, Python language features, History, Entering Expressions into the Interactive Shell, The Integer, Floating-Point, and String Data Types, String Concatenation and Replication, Storing Values in Variables, Your First Program, Dissecting Your Program, Flow control, Boolean Values, Comparison Operators, Boolean Operators, Mixing Boolean and Comparison Operators, Elements of Flow Control, Program Execution, Flow Control Statements, Importing Modules, Ending a Program Early with sys.exit(), Functions, def Statements with Parameters, Return Values and return Statements, The None Value, Keyword Arguments and print(), Local and Global Scope, The global Statement, Exception Handling, A Short Program: Guess the Number Text book 1: Chapters 1–3 Number of Hours:9			
Module-2			
Data Structures: Lists: The List Data Type, Working with Lists Strings: Manipulating Strings, Working with Strings, Useful String Methods Tuples and Dictionaries, basics Using Data Structures to Model Real-World Things, Manipulating Strings. Text book 1: Chapters 4–6 Number of Hours:8			
Module-3			
Pattern Matching with Regular Expressions, Finding Patterns of Text Without Regular Expressions, Finding Patterns of Text with Regular Expressions, More Pattern Matching with Regular Expressions,, The findall() Method, Character Classes, Making Your Own Character Classes, The Caret and Dollar Sign Characters, The Wildcard Character, Review of Regex Symbols. Reading and Writing Files, Files and File Paths, The os.path Module, The File Reading/Writing Process, Saving Variables with the shelve Module, Saving Variables with the pprint. pformat() Function Text book 1: Chapters 7, 8 Number of Hours:8			
Module-4			
Classes and objects: Programmer-defined types, Attributes, Rectangles, Instances as return values, Objects are mutable, Copying, Classes and functions: Time, Pure functions, Modifiers, Prototyping versus planning, Classes and methods: Object-oriented features, Printing objects, Another example, The init method, The __str__ method, Operator overloading, Type-based dispatch, Polymorphism. Text book 2: Chapters 15– 18 Number of Hours:9			
Module-5			
HTTP, The World's simplest Web Browser, Retrieving an image over HTTP, Retrieving web pages with urllib, XML, Parsing XML, Looping through nodes, JSON, What is database?, Database Concepts, Database Browser, Creating a database table, SQL, Spidering Twitter, Basic data modeling, Programming with multiple tables, Three kinds of Keys, JOIN			

Text book 3: Chapters12,13(Upto13.5),15	Number of Hours:8
Suggested Learning Resources: (Text Book/ Reference Book/ Manuals): Text books: 1. Al Sweigart, “Automate the Boring Stuff with Python”,1st Edition, No Starch Press, 2015. (Available under CC-BY-NC-SA license at https://automatetheboringstuff.com/) 2. Allen B Downey, “Think Python: How to Think Like a Computer Scientist”, 2nd Edition, Green Tea Press, 2015. (Available under CC-BY-NC license at http://greenteapress.com/thinkpython2/thinkpython2.pdf) (Chapters 15 - 18) (Download pdf/html files from the above links) 3. Charles R. Severance, “Python for Everybody: Exploring Data Using Python 3”, 1st, Create Space Independent Publishing Platform, 2016	
Web links and Video Lectures (e-Resources): • https://www.youtube.com/watch?v=_xQNeOTRyig • https://www.youtube.com/watch?v=kqtD5dpm9C8	
Teaching-Learning Process (Innovative Delivery Methods): The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes. 1. Lecture method (L) does not mean only the traditional lecture method, but a different type of teaching method may be adopted to develop the outcomes. 2. Show Video/animation films to explain the functioning of various programming concepts. 3. Adopt Problem Based Learning (PBL), which fosters students’ Analytical skills, develop thinking skills such as the ability to evaluate, generalize, and analyse information rather than simply recall it. 4. Show the different ways to solve the same problem and encourage the students to come up with their own creative ways to solve them. 5. Discuss how every concept can be applied to the real world - and when that's possible, it helps improve the students' understanding.	

Analog Electronics and Linear Integrated Circuit Lab			
Course Code	1BECL306	Scheme	2025
Type of Course	PCCL	Semester	III
Teaching Hours/Week (L:T:P)	0:0:2	CIE Marks	50
Total Hours of Pedagogy (L:T:P:SL&TW)	0:0:28:2	SEE Marks	50
Credits	1	Total Marks	100
Examination type (SEE)	Practical	Exam Hours	3
Course outcome (Course Skill Set) At the end of the course, the student will be able to: 1. Apply the knowledge of BJT circuit to determine gain, bandwidth and impedance. 2. Apply the knowledge of FET circuit to determine gain, bandwidth and impedance. 3. Understand the working of oscillators. 4. Realize Op-Amp circuit for the application as voltage follower, integrator, and differentiator. 5. Apply modulation technique.			
Note: 1. The laboratory syllabus consists of PART-A and PART-B. While PART-A has 8 conventional experiments, PART-B has 4 typical open-ended experiments. The maximum marks for the laboratory course are 100. 2. Both PART-A and PART-B are considered for CIE and SEE. 3. Students have answer 1(one) question from PART-A and 1(one) question from PART-B. a. The questions set for SEE shall be from among the experiments under PART-A. It is evaluated for 70 marks out of the maximum 100 marks. b. The open-ended question set for SEE shall be any other open-ended question and not selected from the experiments under PART-A. It shall be evaluated for 30 marks. 4. For Continuous Internal Evaluation (CIE) during the semester, classwork shall typically include open-ended questions from Part-B, along with other similar questions designed to enhance the students' skills.			
PART – A FIXED SET OF EXPERIMENTS			
1. Design and Set-up the BJT Common Emitter Voltage Amplifier with and Without Feedback and Determine the Gain, Bandwidth, Input and Output Impedance. 2. Design and Set-up BJT/ FET Based I) Colpitts Oscillator, II) Crystal Oscillator 3. Realize BJT Darlington Emitter Follower and Determine Gain, Input and Output Impedances. 4. Design 4-Bit R - 2R OP-Amp Digital to Analog Converter: I) For A 4-Bit Binary Input Using Toggle Switches, II) By Generating Digital Inputs Using Mod-16 5. Design and Test the Second Order Active Filters and Plot the Frequency Response: I) Low Pass Filter, II) High Pass Filter 6. Design and Testing of Precision Rectifiers: I) Half-Wave Precision Rectifier, II) Full-Wave Precision Rectifier 7. Design and Test Monostable and Astable Multivibrator Using 555 Timer. 8. Design and Test a Regulated Power Supply Using IC 7805 And IC 7905.			
PART – B OPEN ENDED EXPERIMENTS			

Open-ended experiments are a type of laboratory activity where the outcome is not predetermined and students are given the freedom to explore, design, and conduct the experiment based on the problem statements as per the concepts defined by the course coordinator. It encourages creativity, critical thinking, and inquiry-based learning.
1. Concept 1 Design a FET Voltage Series Feedback Amplifier and Determine the Gain, Input and Output Impedances. 2. Concept 2 Design and Experimental Study of a Class-C Tuned Amplifier 3. Concept 3 Design of a Schmitt Trigger for Given UTP And LTP. 4. Concept 4 Design and Testing of a Sample and Hold Circuit.
Suggested Learning Resources: (Text Book/ Reference Book/ Manuals): Text books: 1. Electronic Devices and Circuits, David A Bell, 5th Edition, Oxford, 2018. 2. Op-amps and Linear Integrated Circuits, Ramakanth A Gayakwad, Pearson Education, 4th Edition. 3. Basic electronics, D.P Kothari and I J Nagrath, McGraw Hill Education Pvt ltd, 2015. 4. Robert L. Boylestad, Louis Nashelsky, "Electronic Devices and Circuit Theory", 11th Edition, PHI, 2016.
Web links and Video Lectures (e-Resources): • http://digimat.in/nptel/courses/video/108102112/L01.html • https://www.youtube.com/playlist?list=PLp6ek2hDcoNDAw1BehPFazZ5ogPV8UIQa
Teaching-Learning Process (Innovative Delivery Methods): The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes. 1. While explaining each experiment, also focus on the application of that particular experiment in the electronics industry 2. In labs utilizing software programming, the emphasis must be on student understanding of the logic (e.g. pseudo-code) of the code, which libraries are required, which important functions calls need to be made and expected output. Faculty should not penalize students for making syntax errors in codes in the write up – however students must be able to debug the errors during program execution 3. Students should not memorize pin diagrams, these must be provided to the student during CIE and SEE 4. For record writing, faculty should ensure students write the main objective, design and procedure in their own words – and do not copy from manuals/online. Also, record writing should involve higher Blooms levels and not just Remember and Understand levels.

Analog Electronic Circuit Simulations Lab			
Course Code	1BECL307A	Scheme	2025
Type of Course	Ability Enhancement Course (Laboratory)	Semester	III
Teaching Hours/Week (L :T : P)	0:0:2	CIE Marks	50
Total Hours of Pedagogy (L :T:P:SL&TW)	0: 0:28 :2	SEE Marks	50
Credits	1	Total Marks	100
Type of Examination	Practical	Exam Hours	3
Course outcome (Course Skill Set) At the end of the course, the student will be able to: 1. Apply the knowledge of BJT circuit to determine gain, bandwidth. 2. Apply the knowledge of FET circuit to determine gain, bandwidth. 3. Gain the knowledge on working of oscillators. 4. Realize Op-Amp circuit for the application as DAC. 5. Realize the working of SCR.			
SET OF EXPERIMENTS Siemens HyperLynx AMS/ PSPICE/ Open-source tool can be used for simulation.			
1. Design and Set-up the BJT Common Emitter Voltage Amplifier with and Without Feedback and Determine the Gain, Bandwidth. 2. Design and Set-up BJT/ FET Based I) Colpitts Oscillator, II) Crystal Oscillator 3. Realize BJT Darlington Emitter Follower and Determine Gain. 4. Design 4-Bit R - 2R OP-Amp Digital to Analog Converter: I) For A 4-Bit Binary Input Using Toggle Switches, II) By Generating Digital Inputs Using Mod-16 5. Design and Test the Second Order Active Filters and Plot the Frequency Response: I) Low Pass Filter, II) High Pass Filter 6. Design and Testing of Precision Rectifiers: I) Half-Wave Precision Rectifier, II) Full-Wave Precision Rectifier 7. Design and Test Monostable and Astable Multivibrator Using 555 Timer. 8. Study the Characteristics of SCR.			
Suggested Learning Resources: (Text Book/ Reference Book/ Manuals): Text books: 1. Electronic Devices and Circuits, David A Bell, 5th Edition, Oxford, 2018. 2. Op-amps and Linear Integrated Circuits, Ramakanth A Gayakwad, Pearson Education, 4th Edition. 3. Basic electronics, D.P Kothari and I J Nagrath, McGraw Hill Education Pvt ltd, 2015. 4. M. H. Rashid, "Power Electronics, Circuits, Devices and Applications," 4th Edition Pearson Prentice Hall, Upper Saddle River, 2018.			
Web links and Video Lectures (e-Resources): • http://digimat.in/nptel/courses/video/108102112/L01.html • https://www.youtube.com/playlist?list=PLp6ek2hDcoNDAw1BehPFazZ5ogPV8UIQa			

Teaching-Learning Process (Innovative Delivery Methods):

The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes.

1. While explaining each experiment, also focus on the application of that particular experiment in the electronics industry
2. In labs utilizing software programming, the emphasis must be on student understanding of the logic (e.g. pseudo-code) of the code, which libraries are required, which important functions calls need to be made and expected output. Faculty should not penalize students for making syntax errors in codes in the write up – however students must be able to debug the errors during program execution
3. Students should not memorize pin diagrams, these must be provided to the student during CIE and SEE
4. For record writing, faculty should ensure students write the main objective, design and procedure in their own words – and do not copy from manuals/online. Also, record writing should involve higher Blooms levels and not just Remember and Understand levels.

Digital Application Lab			
Course Code	1BECL307B	Scheme	2025
Type of Course	Ability Enhancement Course (Laboratory)	Semester	III
Teaching Hours/Week (L :T : P)	0:0:2	CIE Marks	50
Total Hours of Pedagogy (L :T:P:SL&TW)	0: 0:28 :2	SEE Marks	50
Credits	1	Total Marks	100
Type of Examination	Practical	Exam Hours	3
Course outcome (Course Skill Set) At the end of the course, the student will be able to: 1. Understand the basic concepts of digital electronics. 2. Develop an application for domestic appliances. 3. Develop an application for industry control. 4. Develop an application for automotive application.			
SET OF EXPERIMENTS Open-source tool /LabVIEW/ Virtual Lab by IIT Bombay can be used for simulation.			
1. To construct logic circuit of washing machine control using a generalized simulator and verify its output. 2. Conduct an experiment to apply the basic OR gate logic in industrial control process. The user will be able to drag and drop the OR gate on the blank canvas and make input-output connections and verify the truth table. 3. Conduct an experiment to apply a basic NOT gate logic in an 8 -bit one's complement circuit. The user will be able to build, simulate and verify the 8-bit one's complementing circuit using the generalized simulator (a blank canvas with click & place facility for selected gates). 4. To apply the NOT gate logic in an application to indicate that the fuel level in a tank has reached the minimum (reserve) level. The user can compute the output of a given circuit with the help of a generalized simulator. The user will be able to construct, simulate and verify the output of the fuel level application. 5. To study the application of a combination of AND and NOT gates in a seat belt warning indicator. The user will be able to construct, simulate and verify the seat belt warning application using the generalized simulator. Display the output on red LED. 6. Conduct an experiment to apply basic AND gate logic to control the on/off action of car wiper. The user will be able to drag and drop the two-input AND gate on the blank canvas and make input-output connections and verify the truth table. 7. To apply the combination of AND & NOT gates in water level controller application. The user will be able to find the output status of the electronic lock for given combination of inputs. The user will be able to analyse, construct and simulate to verify any combinational logic circuit for a given problem statement using the generalized simulator. 8. To study the application of a combinational electronic lock using basic logic gates. The user will be able to design a 4-digit electronic code lock using two-input basic logic gates. Using the generalized simulator, the user can implement his design and verify it's working. 9. To apply NAND gate logic in a level monitoring system in a chemical plant. Using generalized simulator framework, the user will be able to place a NAND gate, make input-output connections and verify its truth table.			

10. To build automobile alarm application using basic logic gates on a generalized simulator canvas and verify its truth table. 11. To construct a cockpit warning light control system using the generalized simulator. The users will be able to state the output condition for given set of combinations of input. The users will be able to select the correct Boolean expression for given combinational circuit. 12. To construct a half adder circuit and verify its truth table using the generalized simulator framework. 13. Verify Binary to Gray and Gray to Binary conversion using NAND gates only.
Suggested Learning Resources: (Text Book/ Reference Book/ Manuals): Text books: 1. M. Morris Mano, Digital Design, 3rd Edition, Prentice Hall of India Pvt. Ltd., 2003 /Pearson Education (Singapore) Pvt. Ltd., New Delhi, 2003. 2. John F.Wakerly, Digital Design, Fourth Edition, Pearson/PHI, 2006 3. Floyd T.L, Digital Fundamentals , 10/e, Pearson Education, 2011 4. C.H.Roth and L.L.Kimney Fundamentals of Logic Design, 7/e, Cengage Learning, 2013
Web links and Video Lectures (e-Resources): • https://da-iitb.vlabs.ac.in/
Teaching-Learning Process (Innovative Delivery Methods): The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes. 1. While explaining each experiment, also focus on the application of that particular experiment in the electronics industry 2. In labs utilizing software programming, the emphasis must be on student understanding of the logic (e.g. pseudo-code) of the code, which libraries are required, which important functions calls need to be made and expected output. Faculty should not penalize students for making syntax errors in codes in the write up – however students must be able to debug the errors during program execution 3. Students should not memorize pin diagrams, these must be provided to the student during CIE and SEE 4. For record writing, faculty should ensure students write the main objective, design and procedure in their own words – and do not copy from manuals/online. Also, record writing should involve higher Blooms levels and not just Remember and Understand levels.

Introduction to Data Structures and OOPs using Python Lab			
Course Code	1BECL307C	Scheme	2025
Type of Course	Ability Enhancement Course (Laboratory)	Semester	III
Teaching Hours/Week (L :T : P)	0:0:2	CIE Marks	50
Total Hours of Pedagogy (L :T:P:SL&TW)	0:0:28:2	SEE Marks	50
Credits	1	Total Marks	100
Type of Examination	Practical	Exam Hours	3
Course outcome (Course Skill Set) At the end of the course, the student will be able to: 1. Analyse various linear and non-linear data structures. 2. Demonstrate the working nature of different types of data structures and their applications. 3. Use appropriate searching and sorting algorithms for the give scenario. 4. Apply the appropriate data structure for solving real world problems.			
SET OF EXPERIMENTS Any open-source tool can be used.			
1. Develop a program in Python for the following: a) Find factorial of a number. b) Generate Fibonacci series. c) Check prime number.			
2. Develop a program in Python for the following: a) Recursive factorial. b) Recursive Fibonacci. c) Tower of Hanoi simulation (using recursion).			
3. Develop a program in Python to perform operations on strings and files: a) Count vowels, words, and characters. b) Search and replace text in a file. c) Read sensor-like data from a text file.			
4. Implement programs using Python built-in data structures: a) Student record management using dictionaries. b) Remove duplicates using sets. c) Matrix operations using nested lists.			
5. Implement programs using Python for stack operations using lists: a) Push, pop, display. b) Infix to postfix conversion. c) Parenthesis balancing.			
6. Develop programs using Python to implement queue data structures: a) Linear queue b) Circular queue c) Simple task scheduling simulation			

7. Develop programs using Python to implement basic searching and sorting algorithms: - Linear search. - Binary search. - Bubble sort. 8. Develop programs using Python to implement singly linked list operations: - Insert at beginning/end. - Delete a node. - Traverse linked list. 9. Develop Python programs using OOP concepts: - Create a class named circuit_measurement with class data variables Voltage and Current, and methods power_estimation () and resistance_estimation (). Implement the above class in a user program - Create a class named rectangle to polar with data variables real and imag, and methods radius () and angle () (in radians). Use this class to perform rectangular-to-polar coordinate conversion 10. Develop programs using Python to implement inheritance-based applications: - Shape hierarchy (Circle, Rectangle, Triangle) - Employee payroll system. 11. Develop a small real-world application: Digital calculator using tkinter module
Suggested Learning Resources: (Text Book/ Reference Book/ Manuals): Text books: 1. Data Structures and Algorithms with Python by Kent D. Lee and Steve Hubbard. 2. Data structures and algorithms in python by Michael T. Goodrich 3. Problem Solving with Algorithms and Data Structures Using Python by Bradley N Miller and David L Ranum.
Web links and Video Lectures (e-Resources): • https://nptel.ac.in/courses/106106145 • https://onlinecourses.nptel.ac.in/noc20_cs26/announcements?force=true
Teaching-Learning Process (Innovative Delivery Methods): The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes. 1. While explaining each experiment, also focus on the application of that particular experiment in the electronics industry 2. In labs utilizing software programming, the emphasis must be on student understanding of the logic (e.g. pseudo-code) of the code, which libraries are required, which important functions calls need to be made and expected output. Faculty should not penalize students for making syntax errors in codes in the write up – however students must be able to debug the errors during program execution 3. Students should not memorize pin diagrams, these must be provided to the student during CIE and SEE 4. For record writing, faculty should ensure students write the main objective, design and procedure in their own words – and do not copy from manuals/online. Also, record writing should involve higher Blooms levels and not just Remember and Understand levels.

Programming Using Java Lab			
Course Code	1BECL307D	Scheme	2025
Type of Course	Ability Enhancement Course (Laboratory)	Semester	III
Teaching Hours/Week (L :T : P)	0:0: 2	CIE Marks	50
Total Hours of Pedagogy (L :T:P:SL&TW)	0: 0:28:2	SEE Marks	50
Credits	1	Total Marks	100
Type of Examination	Practical	Exam Hours	3
Course outcome (Course Skill Set) At the end of the course, the student will be able to: 1. To learn primitive constructs JAVA programming language. 2. Ability to learn and apply concepts of Java programming, exceptional handling and inheritance. 3. To understand object-oriented programming features of JAVA. 4. To gain knowledge on: packages, multithreaded programming and exceptions.			
SET OF EXPERIMENTS Any open-source tool can be used.			
1. Write a JAVA program that prints all real solutions to the quadratic equation $ax^2+bx+c=0$. Read in a, b, c and use the quadratic formula. 2. Write a JAVA program to implement 555 Timer Frequency Calculator 3. Demonstrate the following operations and sign extension with Java programs (i) << (ii) >> (iii) >>> 4. Create a JAVA class called AC power with the following details as variables within it. Voltage Current Phase angle Write a JAVA program to create power objects and print the voltage, current, active power, apparent power, reactive power, instantaneous power, and power factor of these objects with suitable headings. 5. Write a JAVA program to implement ADC Bit-Resolution Quantizer 6. Design a super class called Staff with details as Staff Id, Name, Phone, Salary. Extend this class by writing three subclasses namely Teaching (domain, publications), Technical (skills), and Contract (period). Write a JAVA program to read and display at least 3 staff objects of all three categories. 7. Write a JAVA Program to Simulate a Multi-Waveform Function Generator using Runtime Polymorphism 8. Write a Java program that implements Quick sort algorithm for sorting a list of names in ascending order. 9. Write a Java program that works as a simple calculator. Use a grid layout to arrange buttons for the digits and for the +, -, *, % operations. Add a text field to display the result. Handle any possible exceptions like divided by zero. 10. Write a Java Program that simulates a Traffic Light. The program lets the use select one of three lights: red, yellow Green with radio buttons. On selecting radio button, an appropriate message with “stop” or “Ready” or “GO” should appear above the button in selected color. Initially, there is no message shown 11. Write a JAVA program to demonstrate Hardware Isolation and Signal Access Control using Java Packages 12. Write a Java program that creates three threads. First thread displays “Good Morning” everyone second, the second thread displays “Hello” every two seconds and the third thread displays “Welcome” every three seconds.			

Suggested Learning Resources: (Text Book/ Reference Book/ Manuals):**Text books:**

1. Java: The Complete Reference, Twelfth Edition, by Herbert Schildt, November 2021, McGraw-Hill, ISBN: 9781260463422
2. Programming with Java, 6th Edition, by E Balagurusamy, Mar-2019, McGraw Hill Education, ISBN: 9789353162337.
3. Thinking in Java, Fourth Edition, by Bruce Eckel, Prentice Hall, 2006 (https://sd.blackball.lv/library/thinking_in_java_4th_edition.pdf)
4. Herbert Schildt, Java The Complete Reference, 7th Edition, Tata McGraw Hill, 2007.

Web links and Video Lectures (e-Resources):

- Java Tutorial: <https://www.geeksforgeeks.org/java/>
- Introduction To Programming In Java (by Evan Jones, Adam Marcus and Eugene Wu): <https://ocw.mit.edu/courses/6-092-introduction-to-programming-in-java-january-iap-2010/>
- Java Tutorial: <https://www.w3schools.com/java/>
- Java Tutorial: <https://www.javatpoint.com/java-tutorial>
- https://onlinecourses.nptel.ac.in/noc22_cs47/preview

Teaching-Learning Process (Innovative Delivery Methods):

The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes.

1. While explaining each experiment, also focus on the application of that particular experiment in the electronics industry
2. In labs utilizing software programming, the emphasis must be on student understanding of the logic (e.g. pseudo-code) of the code, which libraries are required, which important functions calls need to be made and expected output. Faculty should not penalize students for making syntax errors in codes in the write up – however students must be able to debug the errors during program execution
3. Students should not memorize pin diagrams, these must be provided to the student during CIE and SEE
4. For record writing, faculty should ensure students write the main objective, design and procedure in their own words – and do not copy from manuals/online. Also, record writing should involve higher Blooms levels and not just Remember and Understand levels.

Community Project (Project Based Learning) Societal Project

[Common to all disciplines of Engineering]

Course Code	1BCP308	Scheme	2025
Type of Course	SDC	Semester	III
Teaching Hours/Week (L :T : P)	30 (0:0:0)	CIE Marks	50
Total Hours of Pedagogy CI(L :T):LI(P):SL:TW	30 (0:0:0:30)	SEE Marks	50
Credits	1	Total Marks	100
Type of Examination	Practical	Exam Hours	02

Course objective:

Rapid urbanisation, population growth, environmental concerns, energy demand, water scarcity, waste management, transportation issues and rural development challenges have created practical problems affecting communities and public welfare. This course aims to integrate technical knowledge, field exposure and societal engagement so that students identify real community issues, frame problem statements and attempt appropriate, feasible and sustainable engineering responses within the scope of a one-semester project.

Course outcome (Course Skill Set):

At the end of the course, the student will be able to:

1. Identify and analyse societal problems through field interaction and technical assessment.
2. Develop skills in project planning, field survey, data collection, implementation, documentation and reporting.
3. Formulate feasible engineering problem statements and solutions considering practical constraints and sustainability aspects.
4. Apply principles of environmental responsibility, ethics and societal welfare in engineering practice through community-oriented interventions.
5. Exhibit multidisciplinary learning and problem-solving ability based on collected data, user feedback and baseline conditions.
6. Demonstrate teamwork and communication skills while interacting with technical and non-technical stakeholders and reflect on project impact on society.

Restrictions and professional conduct:

Community / Societal Projects shall remain **academic, socially responsible, technically focused, and politically neutral**. Students and faculty shall maintain professionalism, ethical conduct, public respect and institutional integrity while interacting with communities, public groups, organisations and stakeholders.

Students and faculty **shall avoid** the following during project execution:

- Association with political party activities, campaigns or propaganda.
- Participation in activities that may disturb communal harmony, public order or institutional reputation.
- Use of the project platform for personal, political, ideological or commercial promotion.
- Disrespectful, discriminatory or unethical interaction with community members or stakeholders.

Projects must also recognise constraints related to **cost, resources, environment and safety, and stakeholder behaviour/cooperation/acceptance** when planning and implementing activities.

Sustainable Development Goals

The faculty shall encourage students to undertake community projects that align with and contribute towards satisfying the National Board of Accreditation (NBA) Criterion 3.6, namely, Evidence of Addressing Sustainable Development Goals (SDGs), which is a major emphasis of the United Nations (UN). The Sustainable Development Goals (SDGs) are illustrated in the figure shown below:

SUSTAINABLE DEVELOPMENT GOALS



SET OF PROJECT ACTIVITIES (TW/SL = 30 HOURS)

Methods for project identification:

Students may identify the project topic using one or more of the following methods:

- Interaction with local rural or urban stakeholders such as local leaders, corporation/municipality personnel, NGOs, panchayat representatives, etc.
- Review of issues highlighted in print and electronic media, and social networks (WhatsApp,

Facebook, X, Instagram, etc.).

- After selecting a broad topic, students shall collect data, conduct a feasibility survey, estimate timelines and resources, approximate expenditure, and then define the problem statement and objectives clearly.

List of Project Topics and Engineering Outcomes:

The following structured sequence of activities is to be carried out during the semester:

Sl. No.	Project title	Measurable engineering outcome	Mapped POs	SDGs
1	Rural and urban water conservation and management systems	Prepare a site-specific water-saving plan, estimate water demand-supply gap, and propose a feasible conservation intervention with quantified savings.	PO1, PO2, PO3, PO4, PO10, PO11	SDG 6, SDG 11, SDG 12
2	Waste management and recycling	Conduct waste audit, classify waste streams, and design a segregation/recycling workflow with measurable reduction in mixed waste.	PO2, PO3, PO4, PO6, PO8, PO10	SDG 11, SDG 12, SDG 13
Sl. No.	Project title	Measurable engineering outcome	Mapped POs	SDGs

3	Afforestation	Identify a degraded area, prepare plantation layout, and demonstrate survival-rate tracking and maintenance schedule for saplings.	P02, P03, P04, P06, P07, P011	SDG 11, SDG 13, SDG 15
4	Biogas generation and power generation	Estimate organic waste availability, size a small biogas system, and document expected gas output or energy yield.	P01, P02, P03, P04, P05, P010	SDG 7, SDG 11, SDG 12
5	DPR for solar power rural plant	Prepare a DPR with load assessment, panel sizing, cost estimate, and projected energy generation and payback.	P01, P02, P03, P04, P05, P010	SDG 7, SDG 9, SDG 11
6	DPR for electrification of village/tribal community	Map unserved/underserved households, propose electrification layout, and estimate coverage, losses, and implementation cost.	P01, P02, P03, P04, P010, P011	SDG 7, SDG 9, SDG 10
7	DPR for rural drainage system	Survey drainage bottlenecks, develop a drainage layout, and estimate runoff handling capacity and flood-risk reduction.	P01, P02, P03, P04, P06, P010	SDG 6, SDG 9, SDG 11
8	Sustainability and environmental health projects	Assess local environmental health issues and propose interventions with measurable improvement in hygiene, safety, or resource use.	P02, P03, P04, P06, P07, P08	SDG 3, SDG 6, SDG 11, SDG 12
9	Plastic free campaigns	Measure plastic usage before and after awareness intervention and report percentage reduction or behavioural change.	P02, P03, P06, P07, P08, P010	SDG 11, SDG 12, SDG 13
10	Health and nutrition awareness	Conduct baseline and post-program awareness survey and document improvement in awareness score or adoption of healthier practices.	P02, P06, P07, P08, P09, P010	SDG 2, SDG 3, SDG 11
11	Societal projects involving health wellness	Identify a community health need and develop an awareness/service module with measurable participation and feedback outcomes.	P02, P03, P06, P08, P09, P010	SDG 3, SDG 11, SDG 17
12	Air quality monitoring system	Collect air-quality data, compare with reference limits, and present trends along with possible mitigation suggestions.	P02, P03, P04, P05, P06, P010	SDG 3, SDG 11, SDG 13
13	Plastic waste reduction campaign	Track collection, segregation, and replacement actions, and quantify reduction in plastic waste generation or littering.	P02, P03, P06, P07, P08, P010	SDG 11, SDG 12, SDG 13
14	Urban greening projects	Prepare greening plan, identify planting locations, and assess environmental benefits such as shade, aesthetics, or heat mitigation.	P02, P03, P04, P06, P07, P011	SDG 11, SDG 13, SDG 15

Sl. No.	Project title	Measurable engineering outcome	Mapped POs	SDGs
15	Smart agriculture solution	Design a low-cost monitoring or irrigation support solution and estimate improvement in water use efficiency or crop support.	PO1, PO2, PO3, PO4, PO5, PO10	SDG 2, SDG 6, SDG 9
16	Smart irrigation system	Estimate crop water requirement, propose control logic, and quantify water savings compared with conventional irrigation.	PO1, PO2, PO3, PO4, PO5, PO10	SDG 2, SDG 6, SDG 12
17	Youth mentoring programmes	Deliver a structured mentoring activity and measure participation, engagement, and skill/awareness improvement.	PO6, PO8, PO9, PO10, PO11	SDG 4, SDG 10, SDG 17
18	Adult education	Conduct learner-needs assessment, prepare instructional content, and measure improvement in basic literacy or numeracy indicators.	PO6, PO8, PO9, PO10, PO11	SDG 4, SDG 10, SDG 17
19	Food security initiatives / cold storage system	Estimate storage need, design a simple cold-chain concept, and document reduction in spoilage or wastage.	PO1, PO2, PO3, PO4, PO5, PO10	SDG 2, SDG 9, SDG 12
20	Human rights awareness with legal systems	Prepare and deliver an awareness module and assess increase in legal awareness and access-to-rights understanding.	PO6, PO7, PO8, PO9, PO10	SDG 5, SDG 10, SDG 16
21	House tax collection through digital media	Develop a simple digital workflow or awareness support and measure improvement in compliance, convenience, or response time.	PO2, PO3, PO4, PO5, PO10, PO11	SDG 9, SDG 11, SDG 16
22	Use of application / application program	Build or demonstrate a useful community app and measure usability, adoption, or efficiency improvement.	PO1, PO3, PO4, PO5, PO9, PO10	SDG 9, SDG 11, SDG 17
23	Creation of Panchayat library	Plan the library model, catalogue resources, and measure usage, accessibility, or participation outcomes.	PO6, PO8, PO9, PO10, PO11	SDG 4, SDG 10, SDG 11, SDG 17

Suggested Learning Resources:**Textbooks:**

1. Phadke, N. D. Community Engagement and Social Innovation. 1st ed., Oxford University Press, New Delhi.
2. Garg, N. R. Project-Based Learning in Engineering Education. 1st ed., Pearson India Education Services, Noida.
3. Gupta, A. and P. Singh. Service Learning and Community Projects. 1st ed., McGraw-Hill Education (India), New Delhi.
4. Wilson, G. Practical Guide to Community Development Projects. 2nd ed., Routledge, London.

Reference books:

1. NPTEL courses on Project-Based Learning and Social Innovation.
2. SWAYAM courses on community engagement and sustainability.
3. UN Sustainable Development Goals portal: <https://sdgs.un.org>
4. MyGov India and local civic body portals for community initiatives.
5. YouTube resources on design thinking, social innovation, and community service learning.
6. NGO and government field-action case studies related to sanitation, water, health, education, and environment.

Web links and Video Lectures (e-Resources):

- NPTEL courses on Engineering Geology (IIT Roorkee, IIT Kharagpur).
- NPTEL courses on Soil Mechanics and Foundation Engineering (IIT Delhi, IIT Bombay).
- Virtual Labs – Geotechnical Engineering Virtual Lab (IIT Kanpur): <https://vlab.co.in>
- Geological Survey of India (GSI) – Maps, reports, and publications: <https://www.gsi.gov.in>
- YouTube channels: Civil Engineering Essentials, The Engineering Classroom (soil mechanics experiments).
- American Society of Civil Engineers (ASCE) – Geotechnical resources and standards

Teaching Learning Process:

The suggested TLP for Community / Societal Project is as follows;

Week / Phase	Teaching-Learning Activity	Faculty Role	Student Activity	Output / Evidence
1	Course introduction, objectives, ethics, conduct, and assessment briefing	Explain course scope, rules, rubrics, and expected deliverables	Understand course requirements and form teams	Team list, course plan
2	Community problem identification and topic selection	Guide topic framing and feasibility	Identify local issues through discussion and observation	Shortlisted project topic
3	Stakeholder interaction and preliminary survey	Demonstrate survey methods and stakeholder engagement	Collect initial data from community/users	Survey notes, interview record
4	Problem statement and objective	Review and refine problem statement	Prepare problem statement, scope, and	Approved problem statement
	formulation		objectives	
5	Literature/case study review and baseline analysis	Suggest relevant references and examples	Study existing solutions and compare approaches	Baseline analysis summary

Week / Phase	Teaching-Learning Activity	Faculty Role	Student Activity	Output / Evidence
6	Project planning and role assignment	Help teams organize tasks, timeline, and responsibilities	Plan activities, divide roles, and prepare schedule	Work plan / Gantt chart
7	Feasibility analysis and solution brainstorming	Check technical feasibility and practicality	Generate possible engineering solutions	Solution alternatives sheet
8	Selection of final solution and design approach	Review final concept and design logic	Finalize concept, methods, and expected outcome	Concept note / design outline
9	Data processing / calculations / sizing / mapping	Support computations and technical review	Analyze data, perform calculations, prepare maps/charts	Technical analysis sheet
10	Prototype / model / audit / DPR / app development	Monitor progress and correct technical issues	Develop the selected output	Working model / draft DPR / map / app
11	Testing, refinement, and implementation planning	Evaluate technical correctness and improvements	Test prototype or validate proposal with users	Test results / revised output
12	Field implementation / demonstration / awareness activity	Facilitate community interaction and documentation	Conduct implementation or demonstration	Photos, attendance, implementation record
13	Documentation of outcomes, feedback, and impact	Review report structure and completeness	Compile findings, stakeholder feedback, and observations	Draft report
14	Presentation preparation and viva practice	Conduct mock presentation and feedback	Prepare slides/poster and rehearse	Presentation deck
15	Final presentation, submission, and reflection	Evaluate performance using rubrics	Present project, submit report, reflect on learning	Final report, presentation, viva

Assessment structure:

A. Continuous Internal Evaluation (CIE) – 50 marks

- Students, in batches of 3–4, shall undertake at least one project (or a combination of two related activities) aiming, as far as possible, at a logical implemented or implementable outcome.
- CIE will be conducted by a departmental committee (HoD/nominee, guide and one senior faculty) using the following rubrics totalling 50 marks:

CIE pass requirement:

- A student shall secure a minimum of 20 marks out of 50 in CIE to become eligible to appear for SEE.

- Students who fail to earn the required CIE marks may be given additional opportunities before closure of the semester; no re-doing of CIE is permitted after the end of the semester.

B. Semester End Examination (SEE) – 100 marks (scaled to 50)

- Examination duration: 2 hours.
- Maximum examination marks: 100 (entered as 100 in the VTU portal; the ITI-SMU module automatically scales 100 → 50 for result processing).
- The SEE shall be conducted batch-wise by a committee of three faculty: project guide, one faculty from the same department and one from a different department.

SEE pass requirement and overall passing rule:

- For a pass in SEE, a student shall secure minimum 35 marks out of 100.
- For declaration of a pass grade, the sum of CIE marks (out of 50) and scaled SEE marks (out of 50) shall be ≥ 40 ; otherwise, the student is awarded grade F (Fail).

CIE Rubric — Continuous Internal Evaluation (Total 50 marks):

Criterion	Weight (marks)	Excellent (Full marks)	Good	Fair	Needs Improvement
Problem identification & Relevance to society/industry /community	8	Problem is clearly identified with strong evidence of societal relevance and well-defined scope.	Problem is relevant and scope is clear with adequate supporting evidence.	Problem defined but justification or evidence is weak.	Problem unclear or not relevant; little/no supporting evidence.
Formation of problem statement; Engineering approach, innovation & feasibility	10	Problem statement is precise; approach is sound, innovative and clearly feasible.	Problem statement adequate; approach feasible with minor gaps.	Problem statement partial; approach feasibility is marginal.	Problem statement missing or approach not feasible.
Interaction with stakeholders; Data collection, analysis & interpretation	8	Thorough stakeholder engagement; data collection is comprehensive; analysis is well-interpreted.	Good engagement; data and analysis adequate with minor gaps.	Limited engagement; data incomplete and analysis superficial.	Little/no stakeholder contact; insufficient or no data/ analysis.
Project documentation; Clarity, organization & problem-solving demonstrated	10	Documentation is complete, well-organized, analytical, and shows strong problem-solving.	Documentation complete and clear with some analytical content.	Documentation basic with omissions; problem-solving limited.	Documentation missing/poorly organized; no clear problem-solving.
Communication & Presentation skills	6	Presentation is confident, structured, and answers queries fully; visuals effective.	Presentation clear and satisfactory; responds to most queries.	Presentation acceptable but lacks clarity or confidence; weak responses.	Presentation poor or absent; unable to respond.
Contribution & Benefit to society (impact, sustainability)	8	Clear, demonstrable societal benefit and sustainability; realistic path to implementation.	Benefit plausible; sustainability considered with minor gaps.	Limited or unclear societal benefit; sustainability not well addressed.	No evident benefit or sustainability; project not community-oriented.

Notes for CIE rubric:

- Minimum CIE pass requirement: 20/50 to be eligible for SEE. [Use departmental committee (HoD/guide + senior faculty) to apply the rubric and record marks.]

SEE Rubric — Semester End Examination (Enter marks out of 100; scaled to 50 later):

Criterion	Weight (marks)	Excellent (Full marks)	Good	Fair	Needs Improvement
Problem identification & Relevance to society/industry/community	16	Problem and relevance are articulated clearly with strong evidence and context.	Problem relevance clear with reasonable evidence.	Problem stated but evidence/context weak.	Problem poorly defined or irrelevant.
Formation of problem statement; Engineering approach, innovation & feasibility	20	Problem statement is precise; approach demonstrates engineering depth, novelty and feasibility.	Approach is technically sound and feasible with minor limitations.	Approach basic; limited technical depth or feasibility concerns.	No viable engineering approach or infeasible plan.
Interaction with stakeholders; Data collection, analysis & interpretation	16	Excellent stakeholder engagement; rigorous data collection and insightful analysis.	Good engagement and acceptable analysis.	Superficial data collection and limited analysis.	Inadequate or no data; poor analysis.
Project documentation; Clarity, technical rigor & problem-solving demonstrated	20	Report is comprehensive, technically rigorous, and demonstrates systematic problem-solving.	Report is complete and technically correct with minor gaps.	Report lacks technical depth or has notable omissions.	Report missing key elements or technically incorrect.
Communication, Presentation & Viva-voce responses	12	Candidate presents expertly, defends work persuasively and answers all queries correctly.	Good presentation and satisfactory answers to most queries.	Presentation adequate but fails to answer some technical questions.	Poor presentation and unable to defend work.
Contribution & Benefit to society; Impact assessment & sustainability	16	Clear, evidence-based assessment of social impact and Sustainability with scaling recommendations.	Reasonable impact discussion and sustainability considerations.	Limited evaluation of impactor sustainability.	No coherent assessment of impact or sustainability.

SEE pass rule:

- Student must secure at least 35/100 in SEE itself to be eligible for pass consideration.
- Final passing is based on CIE (out of 50) + SEE (scaled to 50) ≥ 40 . Otherwise, grade F (Fail) is awarded.

Suggested implementation notes (for BoS / exam committees):

- Use numerical scores from the rubric and keep brief evaluator comments for each criterion.
- For team projects: record individual contribution within the documentation and adjust individual marks using a peer-evaluation sheet or guide's assessment.
- Maintain evidence (field notes, photos, questionnaires, DPRs, reports) as annexures for SEE moderation.
- Use standardized score sheet (spreadsheet) mapping rubric items to columns for easy aggregation and archival.

Additional Mathematics-1 (Common to all Branches)			
Course Code	1BMATDIP310	Scheme	2025
Type of Course	NCMC	Semester	III
Teaching Hours/Week (L:T:P)	3:0:0	CIE Test Marks	50
Total Hours of Pedagogy (L :T:P:SL&TW)	42:0:0:48	CIE MCQ online Test Marks	50
Credits	PP	Total Marks	100
Examination type (CIE)	Theory	Exam Hours	-
Course outcome (Course Skill Set) At the end of the course, the student will be able to: 1. Provide the requisite and relevant background necessary to understand the other important engineering mathematics courses offered for Engineers and Scientists 2. Apply the knowledge of the co-ordinate system to analyze the polar co-ordinate system. 3. Familiarize the importance of calculus associated with one variable and multivariable for all branches of the engineering. 4. Analyze the importance of the integral calculus and their application to real-world problems. 5. Emphasize the knowledge of the vector calculus to all branches of engineering. CO6: Develop the knowledge of Linear Algebra to solve the system of equations.			
Module-1			
Polar Curves and Curvature: Polar coordinates, Polar curves, angle between the radius vector and the tangent, angle between two curves. Pedal equations. Indeterminate forms: L'Hospital's rule. $\left(\frac{\infty}{\infty}, \infty - \infty, 0^0, 1^\infty\right)$ Number of Hours:8			
Module-2			
Partial Differentiation: Introduction: Partial derivatives, total derivative - differentiation of composite functions, Jacobian and problems. Taylor's and Maclaurin's series expansion for one variable (Statement only) – problems. Number of Hours:9			
Module-3			
Integral Calculus: Statement of reduction formulae for $\sin^n(x)$, $\cos^n(x)$, and $\sin^m(x)$, $\cos^m(x)$, and illustrative problems. Evaluation of double integrals and triple integrals-Simple examples. Applications: Area of the region. Number of Hours:9			
Module-4			
Vector Calculus: Differentiation of vector functions: Velocity and acceleration of a particle moving on a space curve. Scalar and vector point functions. Gradient, Divergence and Curl, Solenoidal and irrotational vector fields-Problems. Number of Hours:8			
Module-5			
Introduction to linear algebra: Elementary row transformation of a matrix, Rank of a matrix. Consistency and Solution of system of linear equations - Gauss-elimination method, Gauss Siedel method. Eigenvalues and Eigenvectors (2X2). Number of Hours:8			
Suggested Learning Resources: (Text Book/ Reference Book/ Manuals): Text books: 1. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 44th Ed., 2021 2. Gilbert Strang, Linear Algebra and its Applications, Cengage Publications, 4th Ed., 2022. 3. Erwin Kreyszig . Advanced Engineering Mathematics, wiley Reference books / Manuals: 1. H K Das and Rajaneesh Varma, Higher Engineering Mathematics, S. Chand. 2. B. V. Ramana, Higher Engineering Mathematics, McGraw Hill.			

3. Mangey Ram, Mathematics in Engineering Sciences, CRC Press 4. N. P. Bali, Manish Goyal - A Textbook of Engineering Mathematics, Laxmi Publications 5. David Poole, Linear Algebra, Cengage Learning India Pvt Ltd.

Web links and Video Lectures (e-Resources):

- NPTEL (National Programme on Technology Enhanced Learning) – Mathematics: <https://nptel.ac.in/>.
- SWAYAM learning modules on Engineering mathematics .
- <https://www.coursera.org/> Mathematics for Engineering Specialization

Teaching-Learning Process (Innovative Delivery Methods):

The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching learning process and facilitate the achievement of course outcomes.

Suggested Innovative Delivery Methods may include (but are not limited to):

- Flipped Classroom
- Problem-Based Learning (PBL)
- Case-Based Teaching
- ICT-Enabled Teaching

Assessment Structure:

For courses offered by the Department of Mathematics, the Continuous Internal Evaluation (CIE) shall be conducted by the concerned course teacher. The CIE may comprise tests, assignments, quizzes, seminars, or other academic activities, as decided by the concerned teacher. The distribution of marks among these activities shall be determined by the course teacher. The total marks awarded for all CIE activities shall be **100 marks**.

The concerned teacher shall upload the CIE marks to the VTU web portal whenever notified by the University.

Online Mode of Offering for 1BMATDIP310 and 1BMATDIP410

The lecture sessions for 1BMATDIP310 and 1BMATDIP410 shall also be made available through the vtu.online.ac.in platform.

1. If all the students admitted to the 3rd and 4th semesters of a college opt to pursue these courses in online mode, the entire course shall be offered online. In such cases, the Centre for Online Education, Mysuru shall conduct the **Continuous Internal Evaluation (CIE) for 100 marks** and submit the CIE marks to the University, with a copy to the concerned college.

2. Within the same college/department, all admitted students shall follow a single mode of learning for these courses. A situation where a part of the students opts for the online mode and the remaining students opt for the offline mode **shall not be permitted**.

3. A college intending to offer these courses through the online mode shall submit the list of eligible students to the Registrar, Visvesvaraya Technological University (VTU), Belagavi, for approval. Only after obtaining the approval of the University shall the students be permitted to enrol for the online course through the vtu.online.ac.in platform. Notwithstanding the above, a student is considered to have **passed the course**, provided the total of **CIE online** examination is at least 40 out of 100 marks.

Mathematics for Machine Learning			
Course Code	1BMATEC401	Scheme	2025
Type of Course	ASC	Semester	IV
Teaching Hours/Week (L:T:P)	3:0:0	CIE Marks	50
Total Hours of Pedagogy per semester L /T/P/SL&TW:	42:0:0:48	SEE Marks	50
Credits	3	Total Marks	100
Examination type (SEE)	Theory	Exam Hours	3
Course outcome (Course Skill Set)			
At the end of the course, the student will be able to:			
1. Apply the concepts of vector spaces, linear transformations, basis, dimension, rank, and nullity to solve problems arising in machine learning and data analysis. 2. Analyze inner product spaces, orthogonality, orthonormal bases, Gram-Schmidt orthogonalization, and QR decomposition for efficient representation of data. 3. Evaluate eigenvalues, eigenvectors, matrix decompositions, positive definite matrices, singular value decomposition, and matrix differentiation techniques used in machine learning algorithms. 4. Apply probability distributions, expectation, variance, covariance, and correlation concepts to model uncertainty and analyze data-driven systems. 5. Utilize sampling theory, hypothesis testing, confidence intervals, stochastic processes, and Markov chains for statistical inference and predictive modeling applications.			
Module-1			
Vector spaces: Definition and examples, subspace, linear span, Linearly independent and dependent sets, Basis and dimension.			
Linear transformations: Definition and examples, Matrix of a linear transformation. Change of bases, Rank and nullity of a linear operator, Rank Nullity theorem.			
Number of Hours:8			
Module-2			
Inner Product Spaces: Inner product, orthogonal sets, angles and orthogonality, orthonormal Basis, orthogonal projections; Gram-Schmidt orthonormalization process; QR- decomposition.			
Number of Hours:8			
Module-3			
Eigen values and Eigenvectors -Diagonalization, Positive definite Matrices and test for positive definiteness, Singular Value Decomposition (2×2).			
Matrix derivatives: Jacobian matrix, Hessian matrix, Gradient of vector and matrix with respect to vector and matrix, pseudo inverse.			
Number of Hours:8			
Module-4			
Review of basic probability theory, Random variables-discrete and continuous Probability distribution function, cumulative distribution function. Mathematical Expectation: mean and variance. Binomial, Poisson, Exponential and Normal distribution– Problems.			
Joint probability distribution: Joint Probability distribution for two discrete random variables, expectation, covariance and correlation.			
Number of Hours:9			
Module-5			
Sampling Theory: Sampling, Sampling distributions, standard error, test of hypothesis for means and proportions, confidence limits for means, student's t-distribution, Chi-square distribution as a test of goodness of fit.			
Stochastic process: Stochastic processes, probability vector, stochastic matrices, fixed probability vector, regular stochastic matrices, Markov chains, higher transition probability-problems.			
Number of Hours:9			

Suggested Learning Resources: (Text Book/ Reference Book/ Manuals):**Text books:**

1. Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong. Mathematics for Machine Learning , Published by Cambridge University Press, Copyright 2020
2. Gilbert Strang, Linear Algebra and its Applications, 3rd edition, Thomson Learning Asia, 2003.
3. B.S. Grewal, Higher Engineering Mathematics, 45th Edition, Khanna Publishers.
4. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley India.
5. H.K. Dass and Er. Rajnish Verma, Higher Engineering Mathematics, S. Chand Publications.

Reference books / Manuals:

1. Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.
2. Trevor Hastie, Robert Tibshirani and Jerome Friedman, The Elements of Statistical Learning, 2nd Edition, Springer, 2009.
3. Sheldon Ross, A First Course in Probability, 10th Edition, Pearson Education, 2019.
4. Kenneth Hoffman and Ray Kunze, Linear Algebra, 2nd Edition, Pearson.
5. David C. Lay, Steven R. Lay and Judi J. McDonald, Linear Algebra and Its Applications, 5th Edition, Pearson.

Web links and Video Lectures (e-Resources):

- <https://nptel.ac.in/courses/111107137>
- NPTEL Courses on Engineering Mathematics
- MIT Open Course Ware Mathematics
- SWAYAM Online Courses

Teaching learning process, assessment structure followed in the course Transform Techniques and Optimization Theory (1BMATEC301) shall be followed.

Applied Computer Organization and Microcontroller			
Course Code	1BEC402	Scheme	2025
Type of course	IPCC	Semester	IV
Teaching Hours/Week (L:T:P)	3:0:2	CIE Marks	50
Total Hours of Pedagogy L:T:P:SL/TW	42:0:28:50	SEE Marks	50
Credits	4	Total Marks	100
Type of Examination (IPCC):	Theory- CIE +SEE, Lab- CIE only.	Exam Hours	3
Course outcome (Course Skill Set) At the end of the course, students will be able to: 1. Analyze computer organization, instruction execution, addressing modes, and evaluate system performance. 2. Evaluate I/O organization, interrupt handling, DMA operations, and cache memory design techniques. 3. Examine processing unit operations and assess pipelining performance and efficiency. 4. Understand and analyze ARM-based embedded system architecture, processor fundamentals, and basic embedded software development. 5. Apply ARM instruction set concepts for embedded programming and for different applications.			
Module-1			
Basic Structure of Computers: Functional Units, Basic Operational Concepts, Bus structure, Performance – Processor Clock, Basic Performance Equation, Clock Rate, Performance Measurement. Machine Instructions and Programs: Memory Location and Addresses, Memory Operations, Instruction and Instruction sequencing, Addressing Modes, Assembler directives Text book-1: 1.2, 1.3, 1.4, 1.6, 2.2, 2.3, 2.4, 2.5, 2.6.1 Number of Hours: 8			
Module-2			
Input/output Organization: Accessing I/O Devices, Interrupts – Interrupt Hardware, Enabling and Disabling Interrupts, Handling Multiple Devices Direct Memory Access: Bus Arbitration, Speed, size and Cost of memory systems. Cache Memories – Mapping Functions. Text book-1: 4.1, 4.2, 4.2.1, 4.2.2, 4.2.3, 4.4, 5.4, 5.5.1 Number of Hours: 8			
Module-3			
Basic Processing Unit: Some Fundamental Concepts: Register Transfers, Performing ALU operations, fetching a word from Memory, Storing a word in memory. Execution of a Complete Instruction. Pipelining: Basic concepts, Role of Cache memory, Pipeline Performance. Text book-1: 7.1, 7.2, 8.1 Number of Hours: 8			
Module-4			
ARM Embedded Systems: Introduction, RISC design philosophy, ARM design philosophy, Embedded system hardware – AMBA bus protocol, ARM bus technology, Memory, Peripherals, Embedded system software – Initialization (BOOT) code, Operating System, Applications. A brief introduction to UART, SPI, and CAN protocols. ARM Processor Fundamentals, ARM core dataflow model, registers, current program status register, Pipeline, Exceptions, Interrupts and Vector Table, Core extensions. Text book-2: 1.1 to 1.5, 2.1 to 2.5 Number of Hours: 10			
Module-5			
Introduction to the ARM Instruction set: Introduction, Data processing instructions, Load – Store instruction, Software interrupt instructions, Program status register instructions, Loading constants, ARMv5E extensions, Conditional Execution. Text book-2: 3.1 to 3.9 Number of Hours: 8			

PRACTICAL COMPONENTS OF IPCC Any open-source tool can be used for simulation.
PRACTICAL COMPONENT OF IPCC 1. Write a program to i) multiply two 16-bit binary numbers. ii) add two 64-bit numbers. 2. Addressing Modes Demonstration (Immediate, Register, Indirect, Indexed) 3. Write a program to find factorial of a number. 4. Write a program to add an array of 16-bit numbers and store the 32-bit result in internal RAM. 5. Write a program to find the square of a number (1 to 10) using look-up table. 6. Write a program to find the largest/smallest number in an array of 32 numbers. 7. Write a program to arrange a series of 32-bit numbers in ascending/descending order. 8. Interface a Stepper motor and rotate it in clockwise and anti-clockwise direction. 9. LED interfacing and Timer Interrupt based LED control 10. Interface a DAC and generate Sine wave and Triangular waveforms. 11. Assembly program to measure instruction execution cycles using a timer/counter. Study the relationship between clock frequency, execution time, and processor performance. 12. Display the Hex digits 0 to F on a 7-segment LED interface, with a suitable delay in between.
Suggested Learning Resources: (Textbook/ Reference Book/ Manuals): Text books: 1. Carl Hamacher, Zvonko Vranesic, Safwat Zaky, Computer Organization, 5 th Edition, Tata McGraw Hill. 2. Andrew N Sloss, Dominic System and Chris Wright, “ARM System Developers Guide”, Elsevier, Morgan Kaufman publisher, 1st Edition, 2008.
Web links and Video Lectures (e-Resources): • https://nptel.ac.in/courses/106106092 • https://nptel.ac.in/courses/106106166 • https://nptel.ac.in/courses/106103180 • https://nptel.ac.in/courses/108105102
Teaching-Learning Process (Innovative Delivery Methods): The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes. 1. Interactive lectures using digital design visualization tools and multimedia presentations to explain logic design concepts. 2. Simulation-based learning using Verilog HDL simulators to model, simulate, and verify digital circuits. 3. Flipped classroom methodology, where students study digital design concepts before class and engage in design exercises during class. 4. Mini-projects and project-based learning involving the design and verification of digital systems such as counters, ALUs, finite state machines, and controllers. 5. Collaborative learning activities through group discussions, peer reviews, and team-based design projects. 6. Continuous assessment through quizzes, Verilog coding challenges, and design reviews to reinforce learning and provide timely feedback.
Questions from the practical component of the IPCC subject shall not appear in the theory SEE component.

Control Systems			
Course Code	1BEC403	Scheme	2025
Type of course	PCC	Semester	IV
Teaching Hours/Week (L:T:P)	3:2:0	CIE Marks	50
Total Hours of Pedagogy L:T:P:SL/TW	42:28:0:50	SEE Marks	50
Credits	4	Total Marks	100
Examination type (SEE)	Theory	Exam Hours	3
Course outcome (Course Skill Set) At the end of the course, students will be able to: 1. Analyze physical systems and their mathematical models using differential equations, transfer functions, and system analogies. 2. Evaluate the effect of feedback on control systems by examining block diagrams, signal flow graphs, and overall system behaviour. 3. Analyze the time response characteristics and steady-state performance of first- and second-order control systems, including the influence of basic controllers. 4. Assess the stability and relative stability of control systems using Routh–Hurwitz criterion and root locus			
Module-1			
Introduction to Control Systems: Types of Control Systems, Effect of Feedback Systems, Differential equation of Physical Systems – Mechanical Systems, Electrical Systems, Analogous Systems. Block diagrams and signal flow graphs: Transfer functions, Block diagram algebra and Signal Flow graphs. Textbook 1: Chapter 1: 1.1,1.3,1.5,1.6, Chapter 2: 2.1,2.2,2.4,2.5,2.6,2.7 Text book 2: Chapter 3 Number of Hours: 14			
Module-2			
Time Response of feedback control systems: Standard test signals, Unit step response of First and Second order Systems. Time response specifications, Time response specifications of second order systems, steady state errors and error constants. Introduction to PI, PD and PID Controllers (excluding design). Textbook 1: Chapter 5:5.1,5.2,5.3,5.4,5.5 Text book 2: Chapter 3 Number of Hours:14			
Module-3			
Stability analysis: Concept of stability, necessary conditions for Stability, Routh- stability criterion, Relative stability criterion, Root-Locus Techniques: Root locus concepts, Construction of root loci, Stability of System. Textbook 1: Chapter 6:6.1,6.2,6.3,6.4,6.5 Chapter 7:7.1,7.2,7.3 Text book 2: Chapter 3 Number of Hours: 14			
Module-4			
Frequency domain analysis and stability using Bode plots: Correlation between time and frequency response, Bode Plots, Experimental determination of transfer function. Textbook 1: Chapter 8: 8.2,8.4,8.5,8.6,8.7 Text book 2: Chapter 3 Number of Hours: 14			
Module-5			
Frequency domain analysis and stability: Introduction to Polar Plots, Mathematical preliminaries, Nyquist Stability criterion, Nyquist plots. Text book 1: Chapter 8: 8.1,8.3, Chapter 9: 9.1,9.2,9.3 Number of Hours:14			

Suggested Learning Resources: (Text Book/ Reference Book/ Manuals):**Text books:**

1. J. Nagarath and MGopa1," ControlSystemsEngineering", NewAge International(p) Limited Publishers, 5 th Edition- 2005, ISBN:81-224-2008-7
2. Analysis and design of Control system using MATLAB by Rao V Dukupati, New Age international (p) Limited Publishers,2006 Publication.

Reference books / Manuals:

1. K.Ogata,"Modern Control Engineering" Pearson .Education Asia I PHI, 4th Edition,2002.
2. K.ChannaVenkatesh and D. Ganesh Rao, "Control Systems", Sanguine TechnicalPublishers.
3. Chi-Tsong Chen, "Analog and Digital Control System Design Transfer-Function, State-Space, and Algebraic Methods", OUP, 2006

Web links and Video Lectures (e-Resources):

- <https://nptel.ac.in/courses/107106081>
- https://onlinecourses.nptel.ac.in/noc24_ee65/preview

Teaching-Learning Process (Innovative Delivery Methods):

The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes.

1. Ensure at least 2 hours of review of mathematical concepts such as matrix algebra, transfer functions, complex variables, first and second order differential equations during the tutorial sessions
2. Present case studies for each module which have industry relevance
3. Use of simulation tools demonstrations in the classroom to illustrate complex concepts

Signals and Systems			
Course Code	1BEC404	Scheme	2025
Type of Course	PCC	Semester	IV
Teaching Hours/Week (L:T:P)	3:0:0	CIE Marks	50
Total Hours of Pedagogy per semester L /T/P/SL&TW:	42:0:0:48	SEE Marks	50
Credits	3	Total Marks	100
Examination type (SEE)	Theory	Exam Hours	3
Course outcome (Course Skill Set) At the end of the course, the student will be able to: 1. Apply knowledge of mathematics and Engineering fundamentals to Identify and characterize different classes of signals. 2. Compute the response and analyze the properties of an LTI system. 3. Analyze the signals in time and frequency domain for non-periodic signals. 4. Apply Fourier representations to analyze sampling process and mixed signals. 5. Apply ZT for broader characterization of discrete time signals and LTI systems.			
Module-1			
Introduction: Definition of signals and systems, Mathematical Representation, Classification of signals, Operation on signals, Elementary signals, Systems viewed as interconnection of operations, Properties of systems. Text book 1: Chapter 1 Number of Hours:8			
Module-2			
Time Domain Representation of LTI Systems: Introduction, Impulse response representation of LTI systems, Properties of impulse response representation, Difference equation representation of LTI systems. Text book 1: Sections 2.1 to 2.3 and 2.4 only Difference equation. Number of Hours:9			
Module-3			
Fourier representations for signals: Introduction, Orthogonality of complex sinusoids, Discrete Time non periodic signals: DTFT representation, Continuous Time non periodic signals: FT representation, Properties of Fourier transforms and DTFT. Text book 1: Sections 3.1 to 3.5 Number of Hours:9			
Module-4			
Applications of Fourier representations: Introduction, Frequency response of LTI systems, Fourier Transform representation of periodic signals, Fourier Transform representation of Discrete time signals, Sampling, reconstruction of continuous time signals. Text book 1: Sections 4.1 to 4.3, 4.5 to 4.6. Number of Hours:8			
Module-5			
Z-Transform: Introduction, Properties of ROC, Properties of Z-transform, inversion of Z-transform, Transform analysis of LTI systems, stability & causality, Unilateral Z-transform and its application. Text book 1: Sections 6.1 to 6.7, 6.10 Number of Hours:8			

Suggested Learning Resources:**Text Books**

1. Simon Haykin, Barry Van Ven, Signals and systems. Ed 2, John Wiley. Indian Ed, 2021.

Reference Book

1. H.P. HSU. Schaum's Outlines- Signals & Systems, Ed 2, TMH, 2010
2. Alan V. Oppenheim, Alan S. Willsky, Syed Hamid Nawab, Signals & systems Ed 2, PHI, 2014

Web links and Video Lectures (e-Resources):

- <https://nptel.ac.in/courses/117101055>
- <https://nptel.ac.in/courses/117104074>

Teaching-Learning Process (Innovative Delivery Methods):

The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes.

1. Lecture method (L) does not mean only the traditional lecture method, but a different type of teaching method may be adopted to develop the outcomes.
2. Show Video/animation films to explain the functioning of various programming concepts.
3. Adopt Problem Based Learning (PBL), which fosters students' Analytical skills, develop thinking skills such as the ability to evaluate, generalize, and analyse information rather than simply recall it.
4. Show the different ways to solve the same problem and encourage the students to come up with their own creative ways to solve them.
5. Discuss how every concept can be applied to the real world - and when that's possible, it helps improve the students' understanding.

Signals and Analog Communications Lab			
Course Code	1BECL405	Scheme	2025
Course Type	PCCL	Semester	IV
Teaching Hours/Week (L:T:P)	0:0:2	CIE Marks	50
Total Hours of Pedagogy (L:T:P:SL&TW)	0:0:28:2	SEE Marks	50
Credits	1	Total Marks	100
Examination type (SEE)	Practical	Exam Hours	3
Course outcome (Course Skill Set) At the end of the course, the student will be able to: 1. Apply knowledge of mathematics and Engineering fundamentals to Identify and characterize different classes of signals. 2. Compute the response and analyze the properties of an LTI system. 3. Analyze the signals in time and frequency domain for non-periodic signals. 4. Understand the basic concepts of AM and FM modulation and demodulation. 5. Design and analyse the electronic circuits for modulation and demodulation circuits.			
Note: 1. The laboratory syllabus consists of PART-A and PART-B. While PART-A has 8 conventional experiments, PART-B has 3 typical open-ended experiments. The maximum marks for the laboratory course are 100. 2. Both PART-A and PART-B are considered for CIE and SEE. 3. Students have answer 1(one) question from PART-A and 1(one) question from PART-B. a. The questions set for SEE shall be from among the experiments under PART-A. It is evaluated for 70 marks out of the maximum 100 marks. b. The open-ended question set for SEE shall be any other open-ended question and not selected from the experiments under PART-A. It shall be evaluated for 30 marks. 4. For Continuous Internal Evaluation (CIE) during the semester, classwork shall typically include open-ended questions from Part-B, along with other similar questions designed to enhance the students' skills.			
PART – A FIXED SET OF EXPERIMENTS Experiments 1 to 4 can be conducted using any open-source tool. Experiments 5 to 10 are to be conducted using discrete components. 1. Generation of Sinusoidal Signal and Energy Computation of Real and Complex Signals. 2. Response Calculation of Discrete LTI System Using Convolution. 3. Finding The DTFT Of Signal $x(n) = \{1, 2, 3, 4\}$ and Compute Its Magnitude and Phase Response. 4. Sampling Theorem Verification and Plotting Spectrum of Discrete Time Signals. 5. Design a Frequency Modulator using VCO and FM Demodulator using PLL. 6. Design and Plot the Frequency Response of Preemphasis and Deemphasis Circuits. 7. Design and Test BJT/FET Mixer. 8. Design and Test Pulse Sampling, Flat Top Sampling and Reconstruction. 9. Design and Test Pulse Amplitude Modulation and Demodulation. 10. Generation and Detection of Pulse Position Modulation and Pulse Width Modulation.			
PART – B OPEN ENDED EXPERIMENTS Open-ended experiments are a type of laboratory activity where the outcome is not predetermined and students are given the freedom to explore, design, and conduct the experiment based on the problem statements as per the concepts defined by the course coordinator. It encourages creativity, critical thinking, and inquiry-based learning.			

Concept 1 and 2 can be conducted using any open-source tool. Concept 3 can be conducted using discrete components.

1. Concept 1 Frequency Domain Representations of Single Tone and Multi Tone Sinusoidal Signals
2. Concept 2 Performing Signal Operations in Time Domain (Time Shifting, Time Scaling, And Time Reversal)
3. Concept 3 Test the Balanced Modulator / Lattice Modulator

Suggested Learning Resources: (Text Book/ Reference Book/ Manuals):

Text books:

1. Simon Haykin, Barry Van Ven, Signals and systems. Ed 2, John Wiley. Indian Ed, 2008.
2. Louis E Frenzel, Principles of Electronic Communication Systems, 3rd Edition, Mc Graw Hill Education (India) Private Limited, 2016. ISBN: 978-0-07-066755-6.

Teaching-Learning Process (Innovative Delivery Methods):

The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes.

1. While explaining each experiment, also focus on the application of that particular experiment in the electronics industry
2. In labs utilizing software programming, the emphasis must be on student understanding of the logic (e.g. pseudo-code) of the code, which libraries are required, which important functions calls need to be made and expected output. Faculty should not penalize students for making syntax errors in codes in the write up – however students must be able to debug the errors during program execution
3. Students should not memorize pin diagrams, these must be provided to the student during CIE and SEE
4. For record writing, faculty should ensure students write the main objective, design and procedure in their own words – and do not copy from manuals/online. Also, record writing should involve higher Blooms levels and not just Remember and Understand levels.

Debugging and Fault Analysis of Analog and Digital Circuits Lab			
Course Code	1BECL406A	Scheme	2025
Type of Course	Ability Enhancement Course (Laboratory)	Semester	IV
Teaching Hours/Week (L :T : P)	0:0:2	CIE Marks	50
Total Hours of Pedagogy (L :T:P:SL&TW)	0:0:28:2	SEE Marks	50
Credits	1	Total Marks	100
Type of Examination	Practical	Exam Hours	3
Course outcome (Course Skill Set) At the end of the course, the student will be able to: 1. Read and analyse datasheets of various analog and digital IC's. 2. Understand the working of various analog and digital equipment's and instruments. 3. Find the faults in various analog and digital equipment's and instruments. 4. Debug faults in various analog and digital equipment's and instruments.			
SET OF EXPERIMENTS			
1. Data Sheet Reading - Understanding the Electrical/ Mechanical /Thermal Characteristics, Absolute Maximum Ratings. 2. Digital Multimeters-Digital Multimeters and LCR Meters- Resolution, Measurements of Voltage (Range), Current (Range), AC, DC, RMS Values, Ave Values, Ripple, Measurements of Frequency, Measurement Of R, Diode, Transistor, SCR, MOSFET Etc, Buzzer Mode, Using Special Probes to Measure Temperature, Excite Few RLC, Fuse Wire Etc with Destructive and Non-Destructive Tests, Along with The Help of Scope and Power Supply. 3. Power Supplies-Setting Up the Power Supply for Output, Voltage Settings, Resolution, Current Limit Setting, Tracking Mode, Float Ground, Common Ground, Over Loading Effect on Power Supplies, Isolated Ground in Dual or Triple Output Power Supplies, Output Noise Measurements, Rise Time, Fall Time and Ripple. 4. Signal Generators - Understanding the Signal Generators - Resolution, Amplitude, Frequency Settings, Waveform Generation, Modulating Two Channels to Get Various Other Kinds of Waveforms from DC To 500khz, Measure the Same with Scope and Confirm the Read Value, Connect A Few RLC Circuits to Observe the Waveform Characteristics. 5. Finding Fault and Debugging 555 Timer-Based Industry Application-Based Circuit 6. Finding Fault and Debugging 12 Volts Fixed and Variable Voltage Power Supply. 7. Finding Fault and Debugging of Domestic Use Stabilizer. 8. Finding Fault in An Inverter Circuit. 9. Debugging Bluetooth Audio Amplifier Board. 10. Debugging of mobile phones. 11. Finding fault in a BJT amplifier Q-Point Drift 12. Finding fault in a combinational stuck at logic probe. 13. Debugging the glitch in the counter. 14. Debugging FM Radio Receiver Board. 15. Finding fault in BS6 automotive sensors using appropriate OBD scanner.			

General instruction regarding fault finding and debugging

1. Open resistor
2. Shorted capacitor
3. Faulty diode
4. Defective transistor
5. Damaged relay
6. Broken PCB track
7. Loose connector
8. Wrong component value
9. Failed voltage regulator
10. Dry solder joint

Suggested Learning Resources: (Text Book/ Reference Book/ Manuals):

Text books:

1. Horowitz, P. and Hill, W., The Art of Electronics, 3rd Ed., Cambridge University Press, 2015
Pease, R.A., Troubleshooting Analog Circuits, 2nd Ed., Newnes/Elsevier, 2013.

Web links and Video Lectures (e-Resources):

Tektronix Inc., *The XYZs of Oscilloscopes*, 7th Edition, Tektronix Publications, Beaverton, USA, 2020, https://download.tek.com/document/03W_8605_7_HR_Letter.pdf (this link is to help faculty and students understand how oscilloscopes function, the different features and how to troubleshoot. The same manual can be used for any brand of oscilloscopes, the principles remain the same)

IOT Lab			
Course Code	1BECL406B	Scheme	2025
Type of Course	Ability Enhancement Course (Laboratory)	Semester	IV
Teaching Hours/Week (L :T : P)	0:0: 2	CIE Marks	50
Total Hours of Pedagogy (L :T:P:SL&TW)	0:0:28:2	SEE Marks	50
Credits	1	Total Marks	100
Type of Examination	Practical	Exam Hours	3
Course outcome (Course Skill Set) At the end of the course, the student will be able to: 1. Explain the Internet of Things and its hardware and software components. 2. Interface I/O devices, sensors & communication modules. 3. Remotely monitor data and control devices. 4. Develop real-life IoT-based projects.			
SET OF EXPERIMENTS Open-source tool can be used for simulation.			
1. (a) To interface LED/Buzzer with Raspberry Pi and write a program to 'turn ON' LED for 1 sec after every 2 seconds. (b) To interface Push button/Digital sensor (IR/LDR) with Raspberry Pi and write a program to 'turn ON' LED when push button is pressed or at sensor detection. (c) To interface motor using relay with Raspberry Pi and write a program to 'turn ON' motor when push button is pressed. 2. (a) To interface DHT11 sensor with Raspberry Pi and write a program to print temperature and humidity readings. (b) To interface OLED with Raspberry Pi and write a program to print temperature and humidity readings on it 3. (a) Write an Arduino/Raspberry Pi program to interface the Soil Moisture Sensor. (b) Write an Arduino/Raspberry Pi program to interface the LDR/Photo Sensor 4. (a) Write a program to interface an Ultrasonic Sensor with Raspberry Pi. (b) Write a program to interface an IR sensor with Raspberry Pi. 5. (a) Write a program on Raspberry Pi to upload temperature and humidity data to thingspeak cloud. (b) Write a program on Raspberry Pi to retrieve temperature and humidity data from thingspeak cloud. 6. Write a program to interface LED using Telegram App 7. (a) Write a program on Raspberry Pi to publish temperature data to MQTT broker. (b) Write a program on Raspberry Pi to subscribe to MQTT broker for temperature data and print it 8. (a) Write a program to create TCP server on Raspberry Pi and respond with temperature and humidity data to TCP client when requested. (b) Write a program to create UDP server on Raspberry Pi and respond with temperature and humidity data to UDP client when requested. 9. (a) Write a program to interface Bluetooth with Raspberry Pi and turn LED ON/OFF when '1'/'0' is received from smartphone using Bluetooth.			

(b) Write a program to interface Bluetooth with Raspberry Pi and send sensor(IR/LDR) data to smartphone using Bluetooth

10. Write a program to interface servo motor to Raspberry Pi using PWM control.

11. Write a program to interface fingerprint sensor with Raspberry Pi to

(a) Enroll a user

(b) Authenticate the user

(c) Delete a user

12. Write a program to interface RFID module with Raspberry Pi to read and write the information.

13. To install MySQL database on Raspberry Pi and perform basic SQL queries.

14. (a) Develop a Python program to interface a temperature sensor with Raspberry Pi and log the temperature readings into an SQLite3 database.

(b) Develop a Python program to retrieve stored temperature readings from an SQLite3 database and print the values on the console.

Suggested Learning Resources: (Text Book/ Reference Book/ Manuals):

Text books:

1. Vijay Madiseti, Arshdeep Bahga, Internet of Things. "A Hands on Approach", University Press.
2. Dr. SRN Reddy, Rachit Thukral and Manasi Mishra, "Introduction to Internet of Things: A practical Approach", ETI Labs

Web links and Video Lectures (e-Resources):

- <https://nptel.ac.in/courses/106105166>

Teaching-Learning Process (Innovative Delivery Methods):

The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes.

1. While explaining each experiment, also focus on the application of that particular experiment in the electronics industry
2. In labs utilizing software programming, the emphasis must be on student understanding of the logic (e.g. pseudo-code) of the code, which libraries are required, which important functions calls need to be made and expected output. Faculty should not penalize students for making syntax errors in codes in the write up – however students must be able to debug the errors during program execution
3. Students should not memorize pin diagrams, these must be provided to the student during CIE and SEE
4. For record writing, faculty should ensure students write the main objective, design and procedure in their own words – and do not copy from manuals/online. Also, record writing should involve higher Blooms levels and not just Remember and Understand levels.

Basics of Machine Learning Lab			
Course Code	1BECL406C	Scheme	2025
Type of Course	Ability Enhancement Course (Laboratory)	Semester	IV
Teaching Hours/Week (L :T : P)	0:0:2	CIE Marks	50
Total Hours of Pedagogy (L :T:P:SL&TW)	0:0:28:2	SEE Marks	50
Credits	1	Total Marks	100
Type of Examination	Practical	Exam Hours	3
Course outcome (Course Skill Set) - At the end of the course, the student will be able to: - Apply machine learning algorithms to real life problems. - Able to make use of different machine learning approaches. - Able to down load different datasets and analysing the signals. - Able to apply different denoising techniques to time series and image signals.			
SET OF EXPERIMENTS Open-source tool can be used for simulation. - Solve the Tic-Tac-Toe problem using the Depth First Search technique. - Show that the 8-puzzle states are divided into two disjoint sets, such that any state is reachable from any other state in the same set, while no state is reachable from any state in the other set. - To represent and evaluate different scenarios using predicate logic and knowledge rules. - To apply the Find-S and Candidate Elimination algorithms to a concept learning task and compare their inductive biases and outputs. - To construct a decision tree using the ID3 or KNN algorithm on a simple classification dataset. - To assess how the ID3 algorithm performs on datasets with varying characteristics and complexity, examining overfitting, underfitting, and decision tree depth. - To examine different types of machine learning approaches (Supervised, Unsupervised, Semi-supervised, and Reinforcement Learning) by setting up a basic classification problem and exploring how each type applies differently. - To understand how Find-S and Candidate Elimination algorithms search through the hypothesis space in concept learning tasks, and to observe the role of inductive bias in shaping the learned concept. - To go through all stages of a real-life machine learning project, from data collection to model fine-tuning, and dimensionality reduction using a regression dataset like the "California Housing Prices." - To perform binary and multiclass classification on the MNIST dataset, analyze performance metrics, and perform error analysis. - Applying different denoising techniques and analyzing the time series signals. - Applying different denoising techniques to images and analyzing them.			

Suggested Learning Resources: (Text Book/ Reference Book/ Manuals):**Text books:**

1. Stuart J. Russell and Peter Norvig , Artificial Intelligence, 3rd Edition, Pearson,2015
2. Elaine Rich, Kevin Knight, Artificial Intelligence, 3rd Edition,Tata McGraw Hill,2013.
3. Tom M. Mitchell, Machine Learning, McGraw-Hill Education, 2013
4. AurelienGeron, Hands-on Machine Learning with Scikit-Learn &Tensor Flow , O'Reilly, Shroff Publishers and Distributors Pvt. Ltd 2019.

Web links and Video Lectures (e-Resources):

- <https://nptel.ac.in/courses/106106139>
- <https://nptel.ac.in/courses/106105152>

Teaching-Learning Process (Innovative Delivery Methods):

The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes.

1. While explaining each experiment, also focus on the application of that particular experiment in the electronics industry
2. In labs utilizing software programming, the emphasis must be on student understanding of the logic (e.g. pseudo-code) of the code, which libraries are required, which important functions calls need to be made and expected output. Faculty should not penalize students for making syntax errors in codes in the write up – however students must be able to debug the errors during program execution
3. Students should not memorize pin diagrams, these must be provided to the student during CIE and SEE
4. For record writing, faculty should ensure students write the main objective, design and procedure in their own words – and do not copy from manuals/online. Also, record writing should involve higher Blooms levels and not just Remember and Understand levels.

Control Systems Lab			
Course Code	1BECL406D	Scheme	2025
Type of Course	Ability Enhancement Course (Laboratory)	Semester	IV
Teaching Hours/Week (L :T : P)	0:0:2	CIE Marks	50
Total Hours of Pedagogy (L :T:P:SL&TW)	0:0:28:2	SEE Marks	50
Credits	1	Total Marks	100
Type of Examination	Practical	Exam Hours	3
Course outcome (Course Skill Set) At the end of the course, the student will be able to: 1. Understand basics of control systems and design mathematical models using block diagram reduction. 2. Understand time domain and frequency domain analysis. 3. Analyze the stability of a system from the transfer function. 4. Develop mathematical model of a motor and generator.			
SET OF EXPERIMENTS Open-source tool /MATLAB/ Scilab/ Octave can be used for simulation.			
1. Implement Block Diagram Reduction Technique and Signal Flow Graph to Obtain Transfer Function a Control System. 2. Simulation of Poles and Zeros of a Transfer Function. 3. Implement Frequency Response of a Second Order System. 4. Implement Frequency Response of a Lead Lag Compensator. 5. Analyze the Stability of the Given System Using Routh Stability Criterion. 6. Analyze the Stability of the Given System Using Root Locus. 7. Analyze the Stability of the Given System Using Bode Plot. 8. Analyze the Stability of the Given System Using Nyquist Plot. 9. Obtain the Time Response from State Model of a System. 10. Stability Analysis of Linear Time Invariant Systems using Matlab 11. Develop a Mathematical Model of DC Motor/ Generator. 12. Develop a Mathematical Model of Servo Motor.			
Suggested Learning Resources: (Text Book/ Reference Book/ Manuals): Text books: 1. L. Nagarath and MGopa1," Control Systems Engineering", NewAge International(p) Limited Publishers, 5 th Edition- 2005, ISBN:81-224-2008-7 2. Analysis and design of Control system using MATLAB by Rao V Dukkipati, New Age international (p) Limited Publishers,2006 Publication.			
Web links and Video Lectures (e-Resources): • https://nptel.ac.in/courses/108106098			
Teaching-Learning Process (Innovative Delivery Methods): The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes. 1. While explaining each experiment, also focus on the application of that particular experiment in the electronics industry 2. Students should not memorize pin diagrams, these can be provided to the student during CIE and SEE			

Biology for Electrical and Electronics Engineers			
Course Code	1BEC407	Scheme	2025
Type of Course	PCC	Semester	IV
Teaching Hours/Week (L:T:P)	2:0:0	CIE Marks	50
Total Hours of Pedagogy (L :T:P:SL&TW)	28:0:0:32	SEE Marks	50
Credits	2	Total Marks	100
Examination type (SEE)	Theory	Exam Hours	3
Course outcome (Course Skill Set) At the end of the course, the student will be able to: 1. Elucidate the basic biological concepts via relevant industrial applications and case studies. 2. Evaluate the principles of design and development, for exploring novel bioengineering projects. 3. Corroborate the concepts of biomimetics for specific requirements. 4. Think critically towards exploring innovative biobased solutions for socially relevant problems.			
Module-1			
Introduction to Biology: The cell (as an information-processing unit): the basic unit of life, Structure and functions of a cell. The Plant Cell and animal cell, Prokaryotic and Eukaryotic cell, Stem cells (inspiration for regenerative bioelectronics) and their application. Importance of special biomolecules: Enzymes, vitamins and hormones. (Classification with one example each, Properties and functions) Number of Hours:5			
Module-2			
Biomolecules and Their Applications (Qualitative): DNA/RNA for bio-computing and molecular electronics, Forensics – DNA fingerprinting, lipids (biodiesel, cleaning agents/detergents), Enzymes (glucose-oxidase in biosensors, lignolytic enzyme in bio-bleaching). Human Blood substitutes - hemoglobin-based oxygen carriers (HBOCs) and perfluorocarbons (PFCs). Number of Hours:5			
Module-3			
Human Organ Systems and Bio Designs (Qualitative): Brain as a CPU system (architecture, CNS and Peripheral Nervous System, signal transmission, EEG, Robotic arms for prosthetics. Engineering solutions for Parkinson's disease). Eye as a Camera system (architecture of rod and cone cells, optical corrections, cataract, lens materials, bionic eye). Heart as a pump system (architecture, electrical signalling - ECG monitoring and heart related issues, reasons for blockages of blood vessels, design of stents, pace makers, defibrillators). Lungs as purification system (architecture, gas exchange mechanisms, spirometry, abnormal lung physiology - COPD, Ventilators, Heart-lung machine). Kidney as a filtration system (architecture, mechanism of filtration, CKD, dialysis systems). Number of Hours:8			
Module-4			
Nature-Bioinspired Materials and Mechanisms (Qualitative): Echolocation (ultrasonography, sonars), Photosynthesis (photovoltaic cells, bionic leaf). Bird flying (GPS and aircrafts), Lotus leaf effect (Super hydrophobic and self-cleaning surfaces), Plant burrs (Velcro), Shark skin (Friction reducing swim suits), Kingfisher beak (Bullet train). Number of Hours:5			
Module-5			
Trends Bioengineering: Bioprinting techniques and materials, 3D printing of ear, bone and skin. 3D printed foods. Electrical tongue and electrical nose in food science, DNA origami and Biocomputing, Bioimaging and Artificial Intelligence for disease diagnosis. Bioremediation and Bio mining. Medical imaging technologies—X-ray, CT, MRI and Ultrasound (overview); Number of Hours:5			

Suggested Learning Resources: (Text Book/ Reference Book/ Manuals):**Text book**

1. Engineering with Biology- Fundamentals and Applications, 2nd Edition, S. Fatima, R. S. Deshpande, M. Ziaullah, Mahantesh Mattada, Santosh Nejekar and Sadashiv Halbhavi, Najekar Publications, 2026

Reference Book

1. Biology for Engineers, Rajendra Singh C and Rathnakar Rao N, Rajendra Singh C and Rathnakar Rao N Publishing, Bengaluru, 2023.
2. Human Physiology, Stuart Fox, Krista Rompolski, McGraw-Hill eBook. 16th Edition, 2022
3. Biology for Engineers, Thyagarajan S., Selvamurugan N., Rajesh M.P., Nazeer R.A., Thilagaraj W., Barathi S., and Jaganthan M.K., Tata McGraw-Hill, New Delhi, 2012.
4. Biology for Engineers, Arthur T. Johnson, CRC Press, Taylor and Francis, 2011 Biomedical Instrumentation, Leslie Cromwell, Prentice Hall 2011.
5. Biology for Engineers, Sohini Singh and Tanu Allen, Vayu Education of India, New Delhi, 2014.
6. Biomimetics: Nature-Based Innovation, Yoseph Bar-Cohen, 1st edition, 2012, CRC Press.
7. Bio-Inspired Artificial Intelligence: Theories, Methods and Technologies, D. Floreano and C. Mattiussi, MIT Press, 2008.
8. Bioremediation of heavy metals: bacterial participation, by C R Sunilkumar, N Geetha A C Udayashankar Lambert Academic Publishing, 2019.
9. 3D Bioprinting: Fundamentals, Principles and Applications by Ibrahim Ozbolat, Academic Press, 2016.
10. Electronic Noses and Tongues in Food Science, Maria Rodriguez Mende, Academic Press, 2016

Web links and Video Lectures (e-Resources):

- <https://nptel.ac.in/courses/121106008>
- <https://freevideolectures.com/course/4877/nptel-biology-engineers-other-non-biologists>
- <https://ocw.mit.edu/courses/20-020-introduction-to-biological-engineering-design-spring-2009>
- <https://ocw.mit.edu/courses/20-010j-introduction-to-bioengineering-be-010j-spring-2006>
- <https://www.coursera.org/courses?query=biology>
- https://onlinecourses.nptel.ac.in/noc19_ge31/preview
- <https://www.classcentral.com/subject/biology>
- <https://www.futurelearn.com/courses/biology-basic-concepts>

Teaching-Learning Process (Innovative Delivery Methods):

The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes.

1. Lecture method (L) does not mean only the traditional lecture method, but a different type of teaching method may be adopted to develop the outcomes.
2. Show Video/animation films to explain the functioning of various programming concepts.
3. Adopt Problem Based Learning (PBL), which fosters students' Analytical skills, develop thinking skills such as the ability to evaluate, generalize, and analyse information rather than simply recall it.
4. Show the different ways to solve the same problem and encourage the students to come up with their own creative ways to solve them.
5. Discuss how every concept can be applied to the real world - and when that's possible, it helps improve the students' understanding.

Environmental Science Project				
Course Code	1BEP408	Scheme	2025	
Type of Course	SDC	Semester	IV	
Teaching Hours/Week (L:T:P)	30 (0:0:0)	CIE Marks	50	
Total Hours of Pedagogy per Semester CI (L /T): LI(P):TW&SL	30 (0:0:0:30)	SEE Marks	50	
Credits	1	Total Marks	100	
Examination type (SEE)	Practical	Exam Hours	03	
Course Objectives: The Environmental Science Project is a practical, applied-learning course that builds students' environmental awareness and responsibility. It focuses on real issues such as climate change, pollution, biodiversity, resource use, soil erosion, and thermal comfort, using field surveys, case studies, data collection, audits, and analysis to develop feasible and sustainable solutions. Students may work individually or in groups of up to four, including interdisciplinary topics, and assessment is based on the prescribed rubrics.				
Course outcome (Course Skill Set): At the end of the course, the student will be able to: 1. Identify and analyse environmental problems through field interaction and technical assessment. 2. Develop skills in project planning, field survey, data collection, implementation, documentation and reporting. 3. Formulate feasible environmental problem statements and solutions considering practical constraints and sustainability aspects. 4. Apply principles of environmental responsibility, ethics and societal welfare in engineering practice through community-oriented interventions. 5. Exhibit multidisciplinary learning and problem-solving ability based on collected data, user feedback and baseline conditions. 6. Demonstrate teamwork and communication skills while interacting with technical and non-technical stakeholders and reflect on project impact on society.				
Restrictions and professional conduct: Environmental Science Projects should be academic, socially responsible, technically focused, and politically neutral. Students and faculty must maintain professionalism, ethical conduct, and respect while working with communities and stakeholders, and avoid any political, ideological, commercial, or disruptive activities. Projects should also consider practical constraints such as cost, resources, safety, environmental impact, and stakeholder cooperation.				
SET OF PROJECT ACTIVITIES (TW/SL = 30 HOURS)				
Methods for project identification: Students may identify the project topic using one or more of the following methods: interaction with local rural or urban stakeholders such as local leaders, corporation/municipality personnel, NGOs, panchayat representatives, etc.; review of issues highlighted in print and electronic media, and social networks; and, after selecting a broad topic, collect data, conduct a feasibility survey, estimate timelines and resources, approximate expenditure, and define the problem statement and objectives clearly.				
List of Project Topics and Engineering Outcomes: The following structured sequence of activities is to be carried out during the semester:				
Sl. No.	Project title	Measurable engineering outcome	Mapped POs	SDGs
1	Rural and urban water conservation and management systems	Prepare a site-specific water-saving plan, estimate water demand–supply gap, and propose a feasible conservation intervention with quantified savings.	PO1, PO2, PO3, PO4, PO10, PO11	SDG 6, SDG 11, SDG 12

2	Waste management and recycling	Conduct waste audit, classify waste streams, and design a segregation/recycling workflow with measurable reduction in mixed waste.	P02, P03, P04, P06, P08, P010	SDG 11, SDG 12, SDG 13
3	Afforestation	Identify a degraded area, prepare plantation layout, and demonstrate survival-rate tracking and maintenance schedule for saplings.	P02, P03, P04, P06, P07, P011	SDG 11, SDG 13, SDG 15
4	Biogas generation and power generation	Estimate organic waste availability, size a small biogas system, and document expected gas output or energy yield.	P01, P02, P03, P04, P05, P010	SDG 7, SDG 11, SDG 12
5	DPR for solar power rural plant	Prepare a DPR with load assessment, panel sizing, cost estimate, and projected energy generation and payback.	P01, P02, P03, P04, P05, P010	SDG 7, SDG 9, SDG 11
6	DPR for electrification of village/tribal community	Map unserved/underserved households, propose electrification layout, and estimate coverage, losses, and implementation cost.	P01, P02, P03, P04, P010, P011	SDG 7, SDG 9, SDG 10
7	DPR for rural drainage system	Survey drainage bottlenecks, develop a drainage layout, and estimate runoff handling capacity and flood-risk reduction.	P01, P02, P03, P04, P06, P010	SDG 6, SDG 9, SDG 11
8	Sustainability and environmental health projects	Assess local environmental health issues and propose interventions with measurable improvement in hygiene, safety, or resource use.	P02, P03, P04, P06, P07, P08	SDG 3, SDG 6, SDG 11, SDG 12
9	Plastic free campaigns	Measure plastic usage before and after awareness intervention and report percentage reduction or behavioural change.	P02, P03, P06, P07, P08, P010	SDG 11, SDG 12, SDG 13
10	Health and nutrition awareness	Conduct baseline and post-program awareness survey and document improvement in awareness score or adoption of healthier practices.	P02, P06, P07, P08, P09, P010	SDG 2, SDG 3, SDG 11
11	Societal projects involving health wellness	Identify a community health need and develop an awareness/service module with measurable participation and feedback outcomes.	P02, P03, P06, P08, P09, P010	SDG 3, SDG 11, SDG 17
12	Air quality monitoring system	Collect air-quality data, compare with reference limits, and present trends along with possible mitigation suggestions.	P02, P03, P04, P05, P06, P010	SDG 3, SDG 11, SDG 13
13	Plastic waste reduction campaign	Track collection, segregation, and replacement actions, and quantify reduction in plastic waste generation or littering.	P02, P03, P06, P07, P08, P010	SDG 11, SDG 12, SDG 13
14	Urban greening projects	Prepare greening plan, identify planting locations, and assess environmental benefits such as shade, aesthetics, or heat mitigation.	P02, P03, P04, P06, P07, P011	SDG 11, SDG 13, SDG 15
15	Smart agriculture solution	Design a low-cost monitoring or irrigation support solution and estimate improvement in water use efficiency or crop support.	P01, P02, P03, P04, P05, P010	SDG 2, SDG 6, SDG 9

16	Smart irrigation system	Estimate crop water requirement, propose control logic, and quantify water savings compared with conventional irrigation.	PO1, PO2, PO3, PO4, PO5, PO10	SDG 2, SDG 6, SDG 12
17	Food security initiatives / cold storage system	Estimate storage need, design a simple cold-chain concept, and document reduction in spoilage or wastage.	PO1, PO2, PO3, PO4, PO5, PO10	SDG 2, SDG 9, SDG 12

Suggested Learning Resources

Particulars	Titles and Publishers
Textbooks	1. Phadke, N. D., Community Engagement and Social Innovation, Oxford University Press, New Delhi. 2. Garg, N. R., Project-Based Learning in Engineering Education, Pearson India Education Services, Noida. 3. Gupta, A. and P. Singh, Service Learning and Community Projects, McGraw-Hill Education (India), New Delhi. 4. Wilson, G., Practical Guide to Community Development Projects, Routledge, London.
Reference books	1. NPTEL courses on Project-Based Learning and Social Innovation. 2. SWAYAM courses on community engagement and sustainability. 3. UN Sustainable Development Goals portal. 4. MyGov India and local civic body portals for community initiatives. 5. YouTube resources on design thinking, social innovation, and community service learning. 6. NGO and government field-action case studies related to sanitation, water, health, education, and environment.
Web links and Video Lectures (e-Resources)	1. NPTEL courses on Engineering Geology. 2. NPTEL courses on Soil Mechanics and Foundation Engineering. 3. Virtual Labs – Geotechnical Engineering Virtual Lab. 4. Geological Survey of India. 5. YouTube channels on civil engineering experiments. 6. American Society of Civil Engineers (ASCE) – Geotechnical resources and standards.

Teaching Learning Process:

The suggested TLP for Community / Societal Project is as follows;

Week / Phase	Teaching-Learning Activity	Faculty Role	Student Activity	Output / Evidence
1	Course introduction, objectives, ethics, conduct, and assessment briefing	Explain course scope, rules, rubrics, and expected deliverables	Understand course requirements and form teams	Team list, course plan
2	Community problem identification and topic selection	Guide topic framing and feasibility	Identify local issues through discussion and observation	Shortlisted project topic
3	Stakeholder interaction and preliminary survey	Demonstrate survey methods and stakeholder engagement	Collect initial data from community/users	Survey notes, interview record
4	Problem statement and objective formulation	Review and refine problem statement	Prepare problem statement, scope, and objectives	Approved problem statement

5	Literature/case study review and baseline analysis	Suggest relevant references and examples	Study existing solutions and compare approaches	Baseline analysis summary
6	Project planning and role assignment	Help teams organize tasks, timeline, and responsibilities	Plan activities, divide roles, and prepare schedule	Work plan / Gantt chart
7	Feasibility analysis and solution brainstorming	Check technical feasibility and practicality	Generate possible engineering solutions	Solution alternatives sheet
8	Selection of final solution and design approach	Review final concept and design logic	Finalize concept, methods, and expected outcome	Concept note / design outline
9	Data processing / calculations / sizing / mapping	Support computations and technical review	Analyze data, perform calculations, prepare maps/charts	Technical analysis sheet
10	Prototype / model / audit / DPR / app development	Monitor progress and correct technical issues	Develop the selected output	Working model / draft DPR / map / app
11	Testing, refinement, and implementation planning	Evaluate technical correctness and improvements	Test prototype or validate proposal with users	Test results / revised output
12	Field implementation / demonstration / awareness activity	Facilitate community interaction and documentation	Conduct implementation or demonstration	Photos, attendance, implementation record
13	Documentation of outcomes, feedback, and impact	Review report structure and completeness	Compile findings, stakeholder feedback, and observations	Draft report
14	Presentation preparation and viva practice	Conduct mock presentation and feedback	Prepare slides/poster and rehearse	Presentation deck
15	Final presentation, submission, and reflection	Evaluate performance using rubrics	Present project, submit report, reflect on learning	Final report, presentation, viva

Assessment structure:

A. Continuous Internal Evaluation (CIE) – 50 marks

- Students, in batches of 3–4, shall undertake at least one project (or a combination of two related activities) aiming, as far as possible, at a logical implemented or implementable outcome.
- CIE will be conducted by a departmental committee (HoD/nominee, guide and one senior faculty) using the following rubrics totalling 50 marks:

CIE pass requirement:

A student shall secure a minimum of 20 marks out of 50 in CIE to become eligible to appear for SEE.

- Students who fail to earn the required CIE marks may be given additional opportunities before closure of the semester; no re-doing of CIE is permitted after the end of the semester.

B. Semester End Examination (SEE) – 100 marks (scaled to 50)

- Examination duration: 2 hours.
- Maximum examination marks: 100 (entered as 100 in the VTU portal; the ITI-SMU module automatically scales 100 → 50 for result processing).
- The SEE shall be conducted batch-wise by a committee of three faculty: project guide, one faculty from the

same department and one from a different department.

SEE pass requirement and overall passing rule:

- For a pass in SEE, a student shall secure minimum 35 marks out of 100.
- For declaration of a pass grade, the sum of CIE marks (out of 50) and scaled SEE marks (out of 50) shall be ≥ 40 ; otherwise, the student is awarded grade F (Fail).

CIE Rubric — Continuous Internal Evaluation (Total 50 marks):

Criterion	Weight (marks)	Excellent (Full marks)	Good	Fair	Needs Improvement
Problem identification & relevance to society/environment	8	Problem clearly identified with strong evidence of relevance and well-defined scope.	Relevant problem with clear scope and adequate evidence.	Problem defined but justification is weak.	Problem unclear or not relevant.
Formation of problem statement; engineering approach, innovation & feasibility	10	Precise statement; sound, innovative, feasible approach.	Adequate statement; feasible with minor gaps.	Partial statement; marginal feasibility.	Missing or infeasible.
Interaction with stakeholders; data collection, analysis & interpretation	8	Thorough engagement; comprehensive data and well-interpreted analysis.	Good engagement; adequate data and analysis.	Limited engagement; incomplete data and superficial analysis.	Little or no stakeholder contact; no data/analysis.
Project documentation; clarity, organization & problem-solving demonstrated	10	Complete, well-organized, analytical, and strong problem-solving.	Complete and clear with some analytical content.	Basic with omissions; problem-solving limited.	Missing/poorly organized; no clear problem-solving.
Communication & presentation skills	6	Confident, structured, and answers queries fully; visuals effective.	Clear and satisfactory; responds to most queries.	Acceptable but lacks clarity or confidence; weak responses.	Poor or absent; unable to respond.
Contribution & benefit to society (impact, sustainability)	8	Clear, demonstrable societal benefit and realistic sustainability path.	Benefit plausible; sustainability considered with minor gaps.	Limited or unclear benefit; sustainability weak.	No evident benefit or sustainability.

Notes for CIE rubric:

- Minimum CIE pass requirement: 20/50 to be eligible for SEE. [Use departmental committee (HoD/guide + senior faculty) to apply the rubric and record marks.]

SEE Rubric — Semester End Examination (Enter marks out of 100; scaled to 50 later):

Criterion	Weight (marks)	Excellent (Full marks)	Good	Fair	Needs Improvement
Problem identification & relevance to society/environment	16	Clearly articulated with strong evidence and context.	Clear with reasonable evidence.	Problem stated but evidence/context weak.	Poorly defined or irrelevant.
Formation of problem statement; engineering approach, innovation & feasibility	20	Precise; demonstrates engineering depth, novelty and feasibility.	Technically sound and feasible with minor limitations.	Basic; limited technical depth or feasibility concerns.	No viable engineering approach or infeasible plan.
Interaction with stakeholders; data collection, analysis & interpretation	16	Excellent engagement; rigorous data collection and insightful analysis.	Good engagement and acceptable analysis.	Superficial data collection and limited analysis.	Inadequate or no data; poor analysis.
Project documentation; clarity, technical rigor & problem-solving demonstrated	20	Comprehensive, technically rigorous, and systematic.	Complete and technically correct with minor gaps.	Lacks technical depth or has notable omissions.	Missing key elements or technically incorrect.
Communication, presentation & viva-voce responses	12	Presents expertly, defends work persuasively and answers all queries correctly.	Good presentation and satisfactory answers to most queries.	Adequate but fails to answer some technical questions.	Poor presentation and unable to defend work.
Contribution & benefit to society; impact assessment & sustainability	16	Clear, evidence-based impact assessment with scaling recommendations.	Reasonable impact discussion and sustainability considerations.	Limited evaluation of impact or sustainability.	No coherent assessment of impact or sustainability.

SEE pass rule:

- Student must secure at least 35/100 in SEE itself to be eligible for pass consideration.
- Final passing is based on CIE (out of 50) + SEE (scaled to 50) ≥ 40 . Otherwise, grade F (Fail) is awarded.

Suggested implementation notes (for BoS / exam committees):

- Use numerical scores from the rubric and keep brief evaluator comments for each criterion.
- For team projects: record individual contribution within the documentation and adjust individual marks using a peer-evaluation sheet or guide's assessment.
- Maintain evidence (field notes, photos, questionnaires, DPRs, reports) as annexures for SEE moderation.
- Use standardized score sheet (spreadsheet) mapping rubric items to columns for easy aggregation and archival.

Awareness Campaign on Single-Use Plastic and E-Waste Management
Suggested Learning Resources: Text Books: - Benny Joseph, Environmental Studies, Tata McGraw-Hill, 2nd Edition. R. Rajagopalan, Environmental Studies: From Crisis to Cure, Oxford University Press, 2005. - Erach Bharucha, Textbook of Environmental Studies, University Press, latest edition as available in library. M. P. Poonia and S. C. Sharma, Environmental Studies, Vikas Publishing House, latest edition as available Reference Books: R. Sivakumar, Principles of Environmental Science and Engineering, Cengage Learning, 2nd Edition. G. Tyler Miller Jr., Environmental Science: Working with the Earth, Thomson Brooks/Cole, 11th Edition. - Pratibha Singh, Anoop Singh, and Piyush Malaviya, Text Book of Environmental and Ecology, Acme Learning, 1st Edition. S. M. Prakash, Environmental Studies, Elite Publishing House, Mangalore, 3rd Edition, 2018.
Web links and Video Lectures (e-Resources): - VTU e-Learning Centre: http://elearning.vtu.ac.in - VTU e-Resources / e-Content: http://elearning.vtu.ac.in/e-Res.php - VTU e-Shikshana Lecture Videos: http://elearning.vtu.ac.in/esk3.php - VTU Consortium Digital Resources: https://access.vtuconsortium.com/msec - VTU e-Learning Centre: http://elearning.vtu.ac.in - VTU e-Resources / e-Content: http://elearning.vtu.ac.in/e-Res.php - VTU e-Shikshana Lecture Videos: http://elearning.vtu.ac.in/esk3.php - VTU Consortium Digital Resources: https://access.vtuconsortium.com/msec
Assessment Structure: Continuous Internal Evaluation (CIE) shall be conducted by the Departmental Project Review Committee consisting of a Senior Professor, the Project Guide, and one additional faculty member nominated by the Head of the Department. Semester End Evaluation (SEE) shall be conducted by university-appointed examiners. The evaluation shall be based on rubrics designed to assess the attainment of NBA-defined graduate attributes, including problem analysis, investigation, design of solutions, communication, teamwork, and professionalism. The weightage of marks for CIE and SEE is 50% each. Pass Criteria: A candidate shall secure a minimum of 40% of the total marks prescribed for Project Work, subject to the required minimum in Continuous Internal Evaluation and Semester End Evaluation as per VTU regulations.

Sample Rubrics for Campus water-quality and development of a water-conservation plan.

Criterion	Weight (%)	4 - Excellent	3 - Good	2 - Satisfactory	1 - Needs improvement
Problem definition and campus context	10	Clearly defines campus water issue, boundaries, water sources, and purpose.	Defines issue and context with minor gaps.	Basic problem stated, but scope is incomplete.	Vague or incorrect problem definition.
Water-quality data collection	15	Uses appropriate sampling points, methods, and frequency; data are well documented.	Methods are mostly appropriate with minor omissions.	Limited methods or incomplete documentation.	Weak or unclear data collection.

Criterion	Weight (%)	4 - Excellent	3 - Good	2 - Satisfactory	1 - Needs improvement
Water-quality parameters and analysis	20	Correctly analyzes key physical, chemical, and microbiological parameters and interprets results.	Analyzes most parameters correctly with minor interpretation gaps.	Some parameters analyzed, but interpretation is weak.	Minimal or incorrect analysis.
Evidence and standards comparison	15	Compares results against relevant standards/guidelines and draws valid conclusions.	Compares most results with reasonable conclusions.	Comparison is partial or uneven.	No meaningful comparison with standards.
Diagnosis of issues and causes	10	Identifies likely causes of poor quality or water loss using evidence.	Identifies main causes with some support.	Causes are mentioned, but weakly supported.	Causes are speculative or absent.
Conservation plan objectives and measures	15	Plan includes clear goals, realistic measures, prioritization, and campus suitability.	Plan is relevant and mostly realistic.	Plan has some useful measures but limited prioritization.	Plan is generic or impractical.
Feasibility and implementation	10	Addresses resources, responsibilities, timeline, and implementation steps clearly.	Covers most implementation aspects.	Implementation is partial or underdeveloped.	Little or no implementation planning.
Cost, savings, and sustainability	5	Provides reasonable cost-benefit thinking and expected savings.	Includes some cost or savings discussion.	Limited financial/savings analysis.	No cost or savings consideration.
Presentation and documentation	0–10	Clear structure, tables/figures, references, and professional report quality.	Well organized with minor issues.	Acceptable but uneven formatting or writing.	Poorly organized and difficult to follow.

Introduction to Analog Communication Systems			
Course Code	1BEC409	Scheme	2025
Type of Course	PCC	Semester	IV
Teaching Hours/Week (L:T:P)	3:0:0	CIE Marks	50
Total Hours of Pedagogy per semester L /T/P/SL&TW:	42:0:0:48	SEE Marks	50
Credits	3	Total Marks	100
Examination type (SEE)	Theory	Exam Hours	3
Course outcome (Course Skill Set) At the end of the course, the student will be able to: 1. Understand and analyse concepts of analog modulation and demodulation schemes. 2. Design and analyse the electronic circuits for modulation and demodulation. 3. Understand the concepts of random variable and random process to model communication systems. 4. Understand and analyse the concepts of digitization of signals.			
Module-1			
Random Variables and Processes: Introduction, Probability, Conditional Probability, Random variables. Statistical Averages: Function of a random variable, Moments, Random Processes, Mean, Correlation and Covariance function: Properties of autocorrelation function, Cross-correlation functions, Gaussian Process: Gaussian Distribution Function. Text book 2: 5.1, 5.2,5.3,5.4,5.5,5.6,5.9 Number of Hours:8			
Module-2			
Amplitude Modulation Fundamentals: AM Concepts, Modulation index and Percentage of Modulation, Sidebands and the frequency domain, AM Power, Single Sideband Modulation. AM Circuits: Amplitude Modulators: Diode Modulator (Exclude Transistor Modulator and Differential Amplifier), collector Modulator. Amplitude Demodulators: Diode Detector (Exclude Crystal Radio Receivers & Synchronous Detection), Balanced Modulators: Lattice Modulators. (Exclude IC Balanced Modulators) Frequency Division Multiplexing: Transmitter-Multiplexer, Receiver-Demultiplexer. Text book 1: 3.1, 3.2,3.3,3.4,3.5,4.2,4.3,4.4,10.2 Number of Hours:9			
Module-3			
Fundamentals of Frequency Modulation: Basic Principles of Frequency Modulation, Principles of Phase Modulation, Modulation index and sidebands, Noise Suppression Effects of FM, Frequency Modulation versus Amplitude Modulation. FM Circuits: Frequency Modulators: Voltage Controlled Oscillators. Frequency Demodulators: Slope Detectors, Phase Locked Loops. Communication Receiver: Super heterodyne receiver, Frequency Conversion: Mixing Principles, Simple diode Mixer. Text book 1: 5.1,5.2,5.3,5.4,5.5,6.1,6.3,9.2,9.3 Number of Hours:8			
Module-4			
Digital Representation of Analog Signals: Introduction, Why Digitize Analog Sources?, The Sampling process, Pulse Amplitude Modulation, Time-Division Multiplexing, Pulse Position Modulation: Generation and Detection of PPM wave. The Quantization Process. Pulse Code Modulation: Sampling, Quantization, Encoding, line Codes, Differential encoding, Regeneration, Decoding, filtering, multiplexing. Text book 2: 7.1,7.2,7.3,7.4,7.5,7.6,7.8,7.9 Number of Hours:9			
Module-5			
Baseband Transmission of Digital signals: Introduction, Inter symbol Interference, Eye Pattern, Nyquist criterion for distortion less Transmission, Baseband M-ary PAM Transmission. Text book 2:8.1,8.4,8.5,8.6,8.7 Noise: Signal to Noise Ratio, External Noise, Internal Noise, Semiconductor Noise, Expressing Noise Levels, Noise in Cascade Stages. Text book 1:9.5 Number of Hours:8			

Suggested Learning Resources:**Text Books**

1. Louis E Frenzel, Principles of Electronic Communication Systems, 3rd Edition, Mc Graw Hill Education (India) Private Limited, 2016. ISBN: 978-0-07-066755-6.
2. Simon Haykin & Michael Moher, Communication Systems, 5th Edition, John Wiley, India Pvt. Ltd, 2010, ISBN: 978-81-265-2151-7.

Reference books / Manuals:

1. B P Lathi, Zhi Ding, "Modern Digital and Analog Communication Systems", Oxford University Press., 4th edition, 2010, ISBN: 97801980738002.
2. Herbert Taub, Donald L Schilling, Goutam Saha, "Principles of Communication systems", 4th Edition, Mc Graw Hill Education (India) Private Limited, 2016. ISBN: 978-1-25-902985-1

Web links and Video Lectures (e-Resources):

- <https://nptel.ac.in/courses/108104091>
- <https://nptel.ac.in/courses/117102059>

Teaching-Learning Process (Innovative Delivery Methods):

The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes.

1. Lecture method (L) does not mean only the traditional lecture method, but a different type of teaching method may be adopted to develop the outcomes.
2. Show Video/animation films to explain the functioning of various analog and digital circuits.
3. Adopt Problem Based Learning (PBL), which fosters students' Analytical skills, develop thinking skills such as the ability to evaluate, generalize, and analyse information rather than simply recall it.
4. Show the different ways to solve the same problem and encourage the students to come up with their own creative ways to solve them.
5. Arrange visits to nearby industries to give brief information about the electronics manufacturing industry.
6. Discuss how every concept can be applied to the real world - and when that's possible, it helps improve the students' understanding.

Additional Mathematics-2 (Common to all Branches)			
Course Code	1BMATDIP410	Scheme	2025
Type of Course	NCMC	Semester	IV
Teaching Hours/Week (L:T:P)	3:0:0	CIE Test Marks	50
Total Hours of Pedagogy (L :T:P:SL&TW)	42:0:0:48	CIE MCQ online Test Marks	50
Credits	PP	Total Marks	100
Examination type (CIE)	Theory	Exam Hours	-
Course outcome (Course Skill Set) At the end of the course, the student will be able to: 1. Provide the requisite and relevant background necessary to understand the other important engineering mathematics courses offered for the Engineers and Scientists. 2. Familiarize the importance of calculus associated with higher order differential equation with constant coefficients for all branches of the engineering. 3. Apply the knowledge to find nonlinear differential equations and their respective solutions. 4. Develop the knowledge of numerical methods to solve equations. 5. Analyze the numerical methods by applying Ordinary Differential Equations. 6. Understand and explain basic probability concepts.			
Module-1			
Ordinary Differential Equations: Higher order ODE's: Linear differential equations of second and higher order equations with constant coefficients. Homogeneous and non-homogeneous equations. Inverse differential operators e^{ax} , $\sin(bx + c)$, $\cos(bx + c)$, and polynomials. Solutions to the initial value problems. Number of Hours:9			
Module-2			
Non-linear differential equations: Non-linear differential equations: Introduction to general and singular solutions, solvable for p. Clairaut's equations, reducible to Clairaut's equations. Number of Hours:8			
Module-3			
Numerical methods-1: Introduction to the numerical methods (All formulae without proof): Finite differences, Problems on Interpolation using Newton's forward and backward difference formulae, Newton's divided difference formula and Lagrange's interpolation formula. Numerical integration: Problems on Trapezoidal, Simpson's (1/3) rd and Simpson's (3/8) th rules. Number of Hours:9			
Module-4			
Numerical Methods-2: Solutions of algebraic and transcendental equations: Problems on Regula-Falsi and NewtonRaphson methods (only formulae). Numerical Solution of Ordinary Differential Equations (ODE's): Numerical solution of ordinary differential equations of first order and first degree - Taylor's series method, Modified Euler's method, Runge-Kutta method of fourth order (No derivations) and problems. Number of Hours:8			
Module-5			
Probability: Introduction. Sample space and events. Axioms of probability. Addition and multiplication theorems. Conditional probability – illustrative examples. Bayes's theorem examples. Number of Hours:8			
Suggested Learning Resources: (Text Book/ Reference Book/ Manuals): Text books: 1. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 44th Ed., 2021 2. M.K. Jain, S.R.K. Iyengar and R.K. Jain, Numerical Methods for Scientific and Engineering Computation, New Age International Publishers, 8thEd., 2022. Reference books / Manuals: 1. H K Das and Rajaneesh Varma, Higher Engineering Mathematics, S. Chand.			

2. B. V. Ramana Higher Engineering Mathematics, McGraw Hill.
3. Mangey Ram, Mathematics in Engineering Sciences, CRC Press
4. N. P. Bali, Manish Goyal - A Textbook of Engineering Mathematics, Laxmi Publications
5. Erwin Kreyszig. Advanced Engineering Mathematics, Wiley.

Web links and Video Lectures (e-Resources):

- NPTEL (National Programme on Technology Enhanced Learning) – Mathematics: <https://nptel.ac.in/>.
- SWAYAM learning modules on Engineering mathematics .
- <https://www.coursera.org/> Mathematics for Engineering Specialization.

Teaching-Learning Process (Innovative Delivery Methods):

The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching learning process and facilitate the achievement of course outcomes.

Suggested Innovative Delivery Methods may include (but are not limited to):

- Flipped Classroom
- Problem-Based Learning (PBL)
- Case-Based Teaching
- ICT-Enabled Teaching

Assessment Structure:

For courses offered by the Department of Mathematics, the Continuous Internal Evaluation (CIE) shall be conducted by the concerned course teacher. The CIE may comprise tests, assignments, quizzes, seminars, or other academic activities, as decided by the concerned teacher. The distribution of marks among these activities shall be determined by the course teacher. The total marks awarded for all CIE activities shall be **100 marks**.

The concerned teacher shall upload the CIE marks to the VTU web portal whenever notified by the University.

Online Mode of Offering for 1BMATDIP310 and 1BMATDIP410

The lecture sessions for 1BMATDIP310 and 1BMATDIP410 shall also be made available through the vtu.online.ac.in platform.

1. If all the students admitted to the 3rd and 4th semesters of a college opt to pursue these courses in online mode, the entire course shall be offered online. In such cases, the Centre for Online Education, Mysuru shall conduct the **Continuous Internal Evaluation (CIE) for 100 marks** and submit the CIE marks to the University, with a copy to the concerned college.

2. Within the same college/department, all admitted students shall follow a single mode of learning for these courses. A situation where a part of the students opts for the online mode and the remaining students opt for the offline mode **shall not be permitted**.

3. A college intending to offer these courses through the online mode shall submit the list of eligible students to the Registrar, Visvesvaraya Technological University (VTU), Belagavi, for approval. Only after obtaining the approval of the University shall the students be permitted to enrol for the online course through the vtu.online.ac.in platform. Notwithstanding the above, a student is considered to have **passed the course**, provided the total of **CIE online** examination is at least 40 out of 100 marks.

Assessment Structure for PCC Courses

Assessment Structure:

The assessment in each PCC course is divided equally between Continuous Internal Evaluation (CIE) and the Semester End Examination (SEE), with each carrying 50% weightage.

- To qualify and become eligible to appear for SEE, in the **CIE**, a student must score at least **40% of 50 marks, i.e., 20 marks.**
- To pass the **SEE**, a student must score at least **35% of 50 marks, i.e., 18 marks.**
- Notwithstanding the above, a student is considered to have **passed the course**, provided the combined total of **CIE and SEE** is at least **40 out of 100 marks.**

Continuous Comprehensive Assessments (CCA)+ Internal Assessment Test (IA) = Continuous Internal Evaluation CIE [25+25=50 marks].

A minimum of two Internal Assessment Tests (IA) shall be conducted, carrying a total of 25 marks.

In addition, Continuous Comprehensive Assessment (CCA) shall be conducted for a total of 25 marks.

It is recommended to include a maximum of two learning activities as part of the CCA to foster the holistic development of students. The suggested activities shall be

Sl. No.	Name of the Learning activity	Maximum Marks
1	Group seminar	15
2	Assignment (at RBL3, RBL4, or RBL5 levels)	10

Rubrics for assessment of these learning activities are provide in the Rubrics Section.

Suggested Learning Activities may include (but are not limited to):

- Course Project
- Case Study Presentation
- Programming Assignment
- Tool/Software Exploration
- Literature Review
- Open Book Test (preferably at RBL4 and RBL5 levels)
- GATE-based Aptitude Test
- Assignment (at RBL3, RBL4, or RBL5 levels)
- Any other relevant and innovative academic activity
- Use of MOOCs and Online Platforms

Suggested Innovative Delivery Methods may include (but are not limited to):

- Flipped Classroom
- Problem-Based Learning (PBL)
- Case-Based Teaching
- Simulation and Virtual Labs
- Partial Delivery of course by Industry expert/ industrial visits
- ICT-Enabled Teaching
- Role Playing

Term Work (TW) and Self Learning (SL) components in Number of Hours per semester		
Term work (includes assignments, seminars, micro projects, industrial visits, any other student activities etc.)		
Sl. No.	Term Work (TW) Activity	Number of Hours / Semester
1	Group seminar	24
2	Assignment	24
OR		
SL: Self Learning, MOOCs, spoken tutorials, online educational resources etc.		
Sl. No.	Self-Learning (SL) Activity	Number of Hours / Semester
1	NPTEL/MooCs course on specific course (faculty have to verify the dates that the NPTEL course is being offered)	48

Continuous Internal Evaluation (CIE)

(i) The CIE marks shall be decided based on internal tests and other outcome-based activities. The continuous internal evaluation shall be carried out by the teacher handling the course.

(ii) Out of 50 marks earmarked for CIE, 25 marks shall be assigned for internal tests. The first test shall be conducted after completing two and half modules of the syllabus and the second one after completing the remaining two and half modules.

A student shall obtain **40% marks or 10 out of 25 marks** allotted for internal tests.

(iii) The remaining 25 marks shall be assigned for Continuous Course Assessment (CCA), conducted as per the rubrics listed in the assessment section of the respective syllabus.

A student shall obtain **40% marks or 10 out of 25 marks** allotted for CCA.

(iv) A student shall obtain a minimum of **40% marks or 20 out of 50 marks** allotted to CIE to become eligible to appear for the SEE.

Passing requirement in SEE

For a pass in the Semester End Examination (SEE), a student shall secure a minimum of 35 marks out of 100.

For the declaration of a pass grade, the sum of the Continuous Internal Evaluation (CIE) marks (out of 50) marks and the SEE marks scaled down to 50 shall be greater than or equal to 40 marks.

If the total (CIE + SEE) is less than 40 marks, the student shall be awarded the grade 'F' (Fail).

Assessment Structure for IPCC Courses

Assessment Structure:

The assessment for each course is equally divided between Continuous Internal Evaluation (CIE) and the Semester End Examination (SEE), with each component carrying **50% weightage** (i.e., 50 marks each). The CIE Theory component will be **25 marks** and CIE Practical component will be **25 marks**.

Questions from the practical component of the IPCC subject shall not appear in the theory SEE component.

CIE Theory component: It consists of IA tests for 25 marks. The CIE Practical component for continuous assessments will be for 15 marks through rubrics and for lab tests will be for 10 marks.

- To qualify and become eligible to appear for SEE, in the **CIE theory component**, a student must score at least **40% of 25 marks, i.e., 10 marks**.
- To qualify and become eligible to appear for SEE, in the **CIE Practical component**, a student must secure a **minimum of 40% of 25 marks, i.e., 10 marks**.
- To pass the **SEE**, a student must secure a **minimum of 35% of 50 marks, i.e., 18 marks**.
- A student is deemed to have **successfully completed the course** if the **combined total of CIE and SEE is at least 40 out of 100 marks**.

CIE Practical Component:

The CIE marks awarded in the case of the Practical component shall be based on the continuous evaluation of the laboratory report using a defined set of rubrics. Each experiment report can be evaluated for 30 marks. The summation of all the experiments marks to be scaled down to 15 marks.

The laboratory test (duration 03 hours) at the end of the last week of the semester /after completion of all the experiments (whichever is early) shall be conducted for 50 marks and scaled down to 10 marks. For laboratory test, the student is required to conduct one experiment each from both Part A and Part B.

Continuous Internal Evaluation (CIE)

(i) Theory component

The CIE marks of 25 shall be earmarked for two tests.

The first test shall be conducted after completing two and half modules of the syllabus and the second one after completing the remaining two and half modules.

Each test shall be conducted for 25 marks. The average of the two tests scaled down to 25 marks shall constitute the test marks for a maximum of 25 marks. The average of the two tests scaled down to 25 marks shall constitute the test marks. For a maximum of 25 marks. **There is no CCA component in IPCC courses**

(ii) Laboratory component

Out of 25 marks: 15 marks shall be assigned for continuous assessments (lab CIE) as per the rubrics listed and 10 marks for the lab test (scaled down to 10 marks). Student shall secure at least **40% of 15 marks, i.e., 6 marks, in the continuous evaluation. Student shall secure at least 40% of 10 marks, i.e., 4 marks, in the lab tests**. The allotment of marks shall be as per the rubrics defined.

To qualify and become eligible to appear for the SEE of the IC/IPCC, a student shall secure at least **40 % of 25 marks, i.e., 10 marks, in the CIE theory component**, and at least **40 % of 25 marks, i.e., 10 marks in the CIE Practical component**.

Table – IPCCs Distribution of marks		
Component	Description	Marks
Allotment of marks for regular class work		
Ability in Conducting Experiments	Initiative, skill, safety practices, teamwork, and independent handling of equipment during regular lab sessions.	5 marks
Laboratory Record / Journal	Regular and neat maintenance of lab record, accuracy of results, and timely submission.	10 marks
Subtotal		15
Allotment of marks for CIE practical test		
Laboratory Test / Experiment Performance	Ability to set up apparatus, follow procedure, take observations, and obtain correct results in the test experiment.	5 marks
Viva-Voce	Understanding of theory, concepts, and experimental procedure; ability to explain results and answer related questions.	5 marks
Subtotal		10
Total		25

Passing Requirements for SEE

For a pass in the Semester End Examination (SEE), a student shall secure a minimum of 35 marks out of 100.

For the declaration of a pass grade, the sum of the Continuous Internal Evaluation (CIE) marks (out of 50) marks and the SEE marks scaled down to 50 shall be greater than or equal to 40 marks.

If the total (CIE + SEE) is less than 40 marks, the student shall be awarded the grade 'F' (Fail).

Assessment Structure for PCC-Lab and AEC Lab Courses

Assessment Structure:

The assessment for each course is equally divided between Continuous Internal Evaluation (CIE) and the Semester End Examination (SEE), with each component carrying **50% weightage** (i.e., 50 marks each).

The CIE marks awarded shall be based on the continuous evaluation of the laboratory report using a defined set of rubrics. Each experiment report can be evaluated for 30 marks. Total marks for the all report shall be scaled down to 30 marks. The laboratory test (duration 03 hours) at the end of the last week of the semester /after completion of all the experiments (whichever is early) shall be conducted for 50 marks and Marks scored shall be scaled down to 20 marks. Final average of marks for report and test shall be out of 50 marks.

For both CIE and SEE, the student is required to conduct one experiment each from both Part A and Part B.

- To qualify and become eligible to appear for SEE, in the **CIE component**, a student must secure a **minimum of 40% of 50 marks, i.e., 20 marks.**
- A student shall obtain **40% marks or 12 out of 30 marks** allotted for continuous evaluation and **40% marks or 08 out of 20 marks** allotted for laboratory test.
- To pass the **SEE component**, a student must secure a **minimum of 35% of 50 marks, i.e., 18 marks.**
- A student is deemed to have **successfully completed the course** if the **combined total of CIE and SEE is at least 40 out of 100 marks**

Term Work (TW) and Self Learning (SL) components in Number of Hours per semester (2) Faculty are suggested to offer only TW and not SL.	
Suggested Term Work (TW) Activity	Number of Hours / Semester
New Experiment to practice on their own	2
OR	
Practice some of the regular experiments with different design requirements on their own	2

Suggested Rubrics for PCC Courses

Note: the rubrics provided below are a reference and can be modified as required.

Rubrics for Learning Activity1: Group seminar (at RBL3, RBL4, or RBL5 levels)

	Superior (5)	Good (4)	Fair (3)	Needs Improvement (2)	Unacceptable (1)
Subject Knowledge (CO/PO Mapping)	Demonstrates deep knowledge of the subject with clear explanation.	Demonstrates adequate knowledge of most topics with clear explanation.	Demonstrates only basic concepts with Superficial knowledge of topic.	Does not have clear understanding of the topic and lacks grasp of information.	No explanation.
Communication Skills (CO/PO Mapping)	Clearly audible by the entire audience.	Audible by most of the audience.	Audience has difficulty to hear and follow the presentation.	Minimal command on vocabulary with inaccurate pronunciation.	Communication characterized by minimal degree of grammatical accuracy with frequent errors.
Viva Voce (CO/PO Mapping)	Able to understand and answer all the questions confidently with relevant explanation.	Able to understand and answer some of the questions with relevant explanation.	Able to answer and explain few questions with less relevance.	Unable to answer most of the questions.	No proper answer.

Rubrics for Learning Activity2: Assignment (at RBL3, RBL4, or RBL5 levels)

	Superior (5)	Good (4)	Fair (3)	Needs Improvement (2)	Unacceptable (1)
Problem understanding (CO/PO Mapping)	Demonstrates complete understanding of the question and assumptions.	Demonstrates good understanding with only minor omissions.	Demonstrates partial understanding with only some unclear assumptions.	Shows limited understanding of the questions.	Weak understanding of the question.
Correct application of theory and correctness of solution (CO/PO Mapping)	Applies all relevant engineering principles, laws, and analytical methods correctly.	Applies most of the relevant engineering principles, laws, and analytical methods correctly.	Applies a few relevant engineering principles, laws, and analytical methods correctly.	Frequent incorrect application of engineering principles and analytical methods.	Major errors in application of engineering principles and analytical methods.

Suggested Rubrics for IPCC courses

Rubrics for CIE Lab test (50marks – later scaled down to 10marks):

Performance Indicators	Excellent	Very Good	Good	Satisfactory
Fundamental Knowledge (10) (P01)	The student has in depth knowledge of the topics related to the course (9-10)	Student has good knowledge of some of the topics related to course (7-9)	Student is capable of narrating the answer but not capable to show in depth knowledge (6-7)	Student has not understood the concepts clearly (5-6)
Design Of Experiment (15) (P02, P03, P05)	Student is capable of discussing more than one design for his/her problem statement and capable of proving the best suitable design with proper reason (14-15)	Student is capable of discussing few designs for his/her problem statement but not capable of selecting best (12-14)	Student is capable of discussing single design with its merits and de-merits (10-12)	Student is capable of explaining the design (9-10)
Implementation with appropriate HW/SW (15) (P03 , P04, P05, P08)	Student is capable of implementing the design with best suitable algorithm/HW considering optimal solution explanation. (14-15)	Student is capable of implementing the design with best suitable algorithm/HW and should be capable of explaining it (12-14)	Student is capable of implementing the algorithm/HW with some explanation. (10-12)	Student is just capable of implementing the algorithm/HW , and unsatisfactory explanation. (8-10)
Result & Analysis (10) (P04, P05, P09)	Student is able to run the program on various cases and compare the result with proper analysis. (10)	Student will be able to run the program for most of the cases and provide proper analysis. (7-9)	Student will be able to run the code for few cases and provide analysis for a few cases. (5-7)	Student will be able to run the program for a simple case but not able to analyze the output. (3-5)

Rubrics for CIE – Continuous assessment (30marks – later scaled to 15 marks):

	Superior	Good	Needs Significant Improvement
Fundamental Knowledge about each experiment (6) (P01)	The student has in depth knowledge of the topics (4-6)	Student has good knowledge of some of the topics (4)	Student has not understood the concepts clearly (2)
Design Of Experiment (8) (P02, P03, P05)	Student capable of discussing more than one design for problem statement and capable of proving the best suitable design with proper reason (6-8)	Student capable of discussing few designs for problem statement , not capable of selecting best design (6)	Student struggles to explain the design (2-4)
Implementation (10) (P03, P04, P05, P08)	Student capable of implementing the design (either alone or in a team) with best suitable algorithm considering optimal solution. (8-10)	Student capable of implementing design (either alone or in a team) with best suitable algorithm and capable of explaining it (8)	Student struggle to implement the design (either alone or in a team) . (2-8)
Quality of Record and Observation write up (6) (P08,P05,P09)	Lab record & observation book are well-organized, with clear sections (4-6)	Lab record/ observation book are organized,, but some sections are not well-defined. (4)	The lab record/ observation book poorly organized, with missing or unclear sections. (2-4)

Note: Term Work and Self Learning components are not included in the assessment. IPCC already has theory and lab assessment in CIE, including a TW/SL component may cause burden on students. Faculty may give 50 hours of SL, however this may not be included for assessment.

Suggested Rubrics for PCC-Lab and AEC Lab Courses

Rubrics for CIE – Continuous Lab assessment (30 marks):

	Superior	Good	Fair	Needs Improvement	Unacceptable
Fundamental Knowledge about each experiment (4) (P01)	The student has in depth knowledge of the topics (4)	Student has good knowledge of some of the topics (3)	Student is capable of narrating the answer but not capable to show in depth knowledge (2)	Student has not understood the concepts clearly (1-2)	Student barely has fundamental knowledge (1)
Design Of Experiment (5) (P02, P03, P05)	Student is capable of discussing more than one design for his/her problem statement and capable of proving the best suitable design with proper reason (5)	Student is capable of discussing few designs for his/her problem statement but not capable of selecting best (4-5)	Student is capable of discussing single design with its merits and de-merits (3-4)	Student struggles to explain the design (2-3)	Student has very weak understanding of the design (1-2)
Implementation (8) (P03, P04, P05, P08)	Student is capable of implementing the design with best suitable algorithm considering optimal solution. (7-8)	Student is capable of implementing the design with best suitable algorithm and should be capable of explaining it (5-6)	Student is capable of implementing the design with proper explanation. (3-4)	Student is capable of implementing the design. (1-2)	Implementation has major errors (1)
Result & Analysis (5) (P04, P05, P09)	Student is able to execute the program/experiment on various cases and compare the result with proper analysis. (4-5)	Student will be able to execute the program/experiment for all the cases. (3-4)	Student will be able to execute the program/experiment for few cases and analyze the output. (2-3)	Student will be able to execute the program/experiment but not able to analyze the output. (1-2)	Program/Experiment execution has important errors and student is incapable to explaining any of them (1)
Quality of Record and Observation write up (8) (P08, P09)	The lab record & observation book are well-organized, with clear sections (e.g., Introduction, Method, Results, Conclusion). (7-8)	The lab record & observation book are organized, with clear sections, but some sections are not well-defined. (5-6)	The lab record & observation book lack clear organization or structure. Some sections are unclear or incomplete. (3-4)	The lab record & observation book are poorly organized, with missing or unclear sections. (1-2)	Student has not submitted lab record or observation book (0)

Rubrics for CIE Lab Test (50 marks, later scaled to 20 marks) and SEE:

Performance Indicators	Excellent	Very Good	Good	Satisfactory
Fundamental Knowledge (5) (P01)	The student has in depth knowledge of the topics related to the course (4)	Student has good knowledge of some of the topics related to course (3)	Student is capable of narrating the answer but not capable to show in depth knowledge (2)	Student has not understood the concepts clearly (1)
Design Of Experiment (5) (P02, P03, P05)	Student is capable of discussing more than one design for his/her problem statement and capable of proving the best suitable design with proper reason (5)	Student is capable of discussing few designs for his/her problem statement but not capable of selecting best (4)	Student is capable of discussing single design with its merits and de-merits (3)	Student is capable of explaining the design (1-2)
Implementation with appropriate HW/SW (5) (P03 , P04, P05, P08)	Student is capable of implementing the design with best suitable algorithm/HW considering optimal solution explanation. (7-8)	Student is capable of implementing the design with best suitable algorithm/HW and should be capable of explaining it (5-6)	Student is capable of implementing the algorithm/HW with some explanation. (3-4)	Student is just capable of implementing the algorithm/H W, and unsatisfactory explanation. (1-2)
Result &Analysis (5) (P04, P05, P09)	Student is able to run the program on various cases and compare the result with proper analysis. (5)	Student will be able to run the program for most of the cases and provide proper analysis. (4)	Student will be able to run the code for few cases and provide analysis for a few cases. (3)	Student will be able to run the program for a simple case but not able to analyze the output. (1-2)

